

Priority Design for Grant-Funding Policies

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Abstract

Grant-making institutions seek to reward merit while guaranteeing support for designated priorities, but unlike seats or slots, grants differ in cost and may be partially funded. We extend the priority-design approach from the allocation of indivisible objects to the division of a fixed pool of funds among such grants. We study two ways of pursuing these priorities: earmarking funds for each category, or setting category-specific floors. With earmarks, we characterize the unique rule that is non-wasteful, respects merit priorities, and splits the cost of a grant eligible for several categories equally among them wherever their reserves allow. With floors, which cannot always be met exactly, we identify the appropriate relaxation and construct a simple rule that satisfies it; under that rule, a grant cannot turn a rejection into an award by requesting more funding at the same merit, and the allocation is unaffected by grants that were never in contention for funding. By contrast, when restricted to integer problems, the rule that funds the highest-merit grants as generously as the constraints allow loses both properties, so efficiency comes at the cost of these guarantees.

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1 Introduction

Scientific research funding in the United States is largely apportioned via centralized funding bodies that must assess applications and partition the resources to contribute to a number of priorities. Primary examples of such institutions are the National Institutes of Health (NIH), the National Science Foundation (NSF) and the Department of Defense (DOD). Each of these institutions makes many details of their procedures publicly available, which we describe in Section 3. We connect their allocation problems to established priority-design schemes: earmarks and floors. **Earmarks**, commonly termed *reserve systems* in the literature, have been proposed and implemented primarily to allocate discrete resources, and often for affirmative-action purposes. These work by setting aside portions of the resource equipped with separate priority orders to ensure adequate consideration of each type of recipient while mitigating justified envy: never assigning a resource to a claimant who is lower priority than someone who did not receive it. Alternatively, type-specific **floors**, or minimum targets, pursue the same goal without partitioning the resource: each type is targeted to receive at least a given amount, and beyond that, all claimants compete under a single priority order. We demonstrate how the same two approaches may be adapted to balance a number of social priorities when apportioning a continuous pool of resources to projects that have differing funding requirements, and allowing for partial funding. In doing so, we introduce an important new application to the priority-design literature.

In our model, applicants make a report, which consists of two numbers, the amount of funding they ask for, and the minimal amount they would accept, and a set of eligible funding categories. Given the report, peer review determines a merit score. The combination of the report and the merit score is called a *grant*, which is what the designer observes; they can choose to reject a grant, or fund it at any level between the minimal level and the full request. There is a set of constraint values for each category which sum to at most the total budget. When the constraints are earmarks, we find a unique characterization after imposing, following [Delacrétaz \(2021\)](#), that categories split the cost of grants evenly: all choice rules that are non-wasteful, respect priorities, and are eligibility-compliant and category-neutral are equivalent to the serial-simultaneous reserve (SR) choice rule (Theorem 1). Moreover, the serial-SR choice rule never penalizes a grant for a weak improvement; if a grant deviates to become cheaper, higher merit, and eligible for more categories, this cannot hurt its prospects of attaining funding (Proposition 2). Finally, if a grant that is worse than all funded grants is added to the problem, the funding assigned to the existing grants is unaffected; the only possible difference is that leftover funding is assigned to the new grant (Proposition 3). We show the serial-SR choice rule operates in polynomial

time (Proposition 4).

When the constraints are floors, we develop a relaxation of strict funding floors, feasible floors. An allocation satisfies feasible floors if for all categories whose floor is unmet, it is not possible to add funds to the highest merit eligible grant without exceeding the floor. In other words, a feasible floor is met if it is not possible to add funds to the next-best grant in the category without over-shooting. Our main result is a constructive choice rule that satisfies the axioms; the floors-first choice rule is non-wasteful, respects priorities, and satisfies feasible floors; furthermore, it is monotonic in cost and independent of non-contending grants (Theorem 2). The significance of cost-monotonicity is that no grant can receive funding for a more costly request compared to a cheaper request, holding merit and eligible categories fixed. Independence of non-contending grants means that the addition of a grant which is lower merit than all funded grants and is not the next-best for some category cannot affect the allocation, unless it is cheap enough to absorb idle funding. In contrast, if we required a stronger notion of floors, strong feasible floors, which is satisfied when the addition of funds towards any eligible grant would lead to over-shooting the floor, the choice rule would fail independence of non-contending grants (Section A.4). This result clarifies the appropriate relaxation of floors on both normative and operational grounds. Normatively, under strong feasible floors a cheap, low-merit application can turn a satisfied floor into an unmet one and pull funds away from higher-merit grants; operationally, whether a floor is met then depends on the cheapest eligible grant rather than the next-best one, so there is no clear margin at which the funder decides. While the floors-first choice rule is monotonic only in cost, the reason it fails type-monotonicity is that the satisfaction of floors may be determined by marginal grants, which are the next-best for the category; under the right circumstances, these marginal grants are better off if they have a lower type (Corollary 1). However, the floors-first choice rule is monotonic in types for non-marginal grants (Proposition 5). There is only one marginal grant per category, so we are able to bound the set of grants which can benefit from lowering their type by the number of categories (Corollary 2). Therefore, in a large population of grants, the share of applicants who are exposed to non-monotonicity in types vanishes. Finally, the floors-first choice rule operates in polynomial time (Proposition 6).

Since the floors-first choice rule does not characterize all choice rules that are non-wasteful, respect priorities and satisfy feasible floors, we turn to a refinement of this set of choice rules. We focus on *integer floor problems*, floor problems whose numeric values are all integers, and *integer-valued choice rules*, choice rules that are defined on integer floor problems and assign integer values of funding. The floors-first choice rule is an integer-valued choice rule when restricted to integer floor problems (Proposition 8). We call the *merit-maximal choice rule* the integer allocation satisfying the axioms that gives the most funding to the highest-merit grant, then, among

those, the most to the second-highest-merit grant, and so on. In other words, it is constrained lexicographically efficient. It is unique (Proposition 9), but it fails independence of non-contending grants, it is not monotonic in cost, merit, or categories, and it is non-monotonic in types for non-marginal grants (Theorem 3). Therefore, the merit-maximal choice rule fails the incentive criteria along all dimensions, and it is sensitive to the addition of grants which should not be in contention for funding. We use this as evidence that a non-efficient choice rule is a better approach for floor problems, which supports use of the floors-first choice rule.

The chief contribution of this paper is to extend a priority-design and reserve-design framework to the context of grant-funding, and more broadly any context which determines how to allocate funds to achieve a number of social objectives. Whereas prior work has largely focused on allocation of discrete resources, we characterize allocation of a pool of funding under two realistic paradigms. As such, the framework has relevance to an array of applications, such as determining financial aid and scholarships, government budgeting, and disaster relief triage.

The paper proceeds as follows. Section 2 reviews the related literature. Section 3 describes the review and funding procedures of the DOD, NSF and NIH, which motivate the two constraint schemes we study. Section 4 establishes the formal model and desiderata. Section 5 treats earmark problems, Section 6 treats floor problems, and Section 7 studies the merit-maximal choice rule for integer floor problems and shows that its efficiency comes at the cost of the incentive and independence properties satisfied by the floors-first choice rule. Section 8 concludes. Proofs are in Appendix A.

2 Related Literature

This paper considers two possible schemes to ensure resources are allocated (or matched) to recipients in a way that respects priorities, while also meeting certain requirements on the distribution of outcomes. There is a deep literature studying this type of question in the context of the college admissions problem and related matching problems which provides an array of approaches to achieve this goal. [Abdulkadiroğlu \(2005\)](#) and [Abdulkadiroğlu and Sönmez \(2003\)](#) introduce type-specific upper quotas or ceilings to the college admissions problem. The former paper considers the college admissions problem, while the latter introduces this approach to school choice. Since then, a number of papers consider different schemes, including reserves, floors, and quotas, to ensure prioritization of each category of recipient in a matching framework.

A long-standing observation about priority-design problems is that allocation decisions are not and should not be determined by lining up candidates according to merit and picking the top n candidates. Rather, a number of distributional objectives are taken into consideration to make a balanced selection ([Alkan and Gale 2003](#),

[Abdulkadiroğlu 2005](#)). Likewise, in a grant-funding problem, there is a merit-score determined by peer review, and funding decisions are made with consideration of this score as well as institutional objectives to create a balanced portfolio of grants. We refer to this process—peer review, then portfolio balance—as “two-tiered” review, following the convention from the Department of Defense ([Department of Defense, 2023](#)).

[Hafalir, Yenmez, and Yildirim \(2013\)](#) study the school choice problem when there are majority quotas versus minority reserves. They are the first to introduce minimum-guarantee reserves, where at each school, students belonging to the affirmative-action category get priority over a portion of school seats until enough students of the category receive a seat; after that point, the seats have a priority order that is not affected by category-inclusion. They find that minority reserves produce an outcome that Pareto-dominates the outcome under majority quotas. They also find that minority reserves produce more favorable outcomes in a simulation exercise. However, we do not propose using a soft reserve system; instead we focus on a feasible version of hard floors, and a hard (set-aside) reserve system separately, allowing us to focus on the new considerations of a grant-funding problem.

[Ehlers, Hafalir, Yenmez, and Yildirim \(2014\)](#) consider the school choice problem with type-specific lower and upper bounds (floors and ceilings, respectively) at each school. They consider an assignment feasible if it assigns every student a seat and it meets the floors and ceilings. They establish limitations on the possibility of feasible assignments that are fair across types, as well as assignments that are fair for same types and non-wasteful. They propose using soft quotas instead, and they propose the *student exchange algorithm*, which is fair for same types and constrained non-wasteful as well as constrained efficient. A notable feature of their choice rule is that, at each school, it dynamically gives high priority for students belonging to a type that has not reached the floor, medium priority when the floor is met but the ceiling is not exceeded, and low priority when the ceiling is exceeded. Our treatment of floors is related: our proposed choice rule prioritizes floors until they are met, then we consider assignment of the remaining funds. As in their setting, hard floors cannot always be met, so we relax them to feasible floors; the floors-first choice rule then respects merit priorities and satisfies feasible floors while remaining non-wasteful (Theorem 2).

A number of papers have studied matching with distributional constraints such as floors and ceilings for each category of student. [Kamada and Kojima \(2015\)](#) develop a choice rule that addresses the rural hospital theorem, namely by reducing the capacity of hospitals in certain regions and re-running deferred acceptance (DA) until the outcome satisfies a desired number of doctors matched to each hospital region. [Fragiadakis and Troyan \(2017\)](#) discuss an improvement on previous solutions to this type of matching problem. They propose a solution that Pareto-dominates the Artificial Caps Deferred Acceptance (ACDA) mechanism, which is the name they give

the mechanism from [Kamada and Kojima \(2015\)](#). Namely, the improved algorithm works by iteratively lowering the overall quotas and type-specific quotas according to a sequence, each time checking whether floors and quotas are met—if so, the procedure stops.

Axiomatic methods entered school choice at the level of the mechanism: [Kojima and Ünver \(2014\)](#) characterize the Boston mechanism by axioms that include conditions on priorities. [Echenique and Yenmez \(2015\)](#) are the first to apply this approach to a school’s choice rule under diversity constraints, showing how real-world constraints pin down a unique choice rule. Our solution to the grant-funding problem with earmarks is an axiomatic characterization, while in the floor problem we characterize a rule satisfying the axioms, and we apply a total order to find the maximum of the set we show is nonempty.

Later, a few papers discuss school-choice problems where students may be eligible for multiple categories. For instance, [Kurata, Hamada, Iwasaki, and Yokoo \(2017\)](#) are the first to give treatment to this problem, while [Sönmez and Yenmez \(2019b\)](#) follow by considering two possible treatments of “overlapping reserves.” In the school choice context, this would mean that they consider a case where students are eligible for multiple reserve categories. They consider two possible treatments, one-to-one and one-to-all. In the first case, a student will satisfy only one reserve slot, and in the latter case, a student satisfies reserve slots for all of her eligible categories. We make use of the one-to-one approach for earmark problems, and the one-to-all approach for floor problems.

[Dur, Kominers, Pathak, and Sönmez \(2018\)](#) show the significance of the order in which reserves are processed. Since then, some more recent papers have considered choice rules that do not depend on an ordering of categories, for instance [Delacrétaz \(2021\)](#). We follow suit by proposing solutions which do not depend on category orderings unless desired.

More recent papers have explored additional applications of matching theory, especially related to triage, using schemes that build on [Hafalir et al. \(2013\)](#). For instance, [Pathak, Sönmez, Ünver, and Yenmez \(2024\)](#) consider the allocation of antiviral treatment and other scarce medical resources. They propose a reserve-system approach. In addition, [Sönmez and Yenmez \(2022\)](#) use matching theory to describe the court-mandated job assignment rule for public jobs in India with an axiomatic characterization. [Sönmez and Yenmez \(2019a\)](#), an earlier version of [Sönmez and Yenmez \(2024\)](#), implicitly use a merit total-order to characterize a unique optimal choice rule. The published version was rewritten and no longer contains this construction; we adopt the idea and give it an explicit treatment in our setting. [Delacrétaz, Kominers, and Teytelboym \(2023\)](#) apply matching theory to refugee resettlement, where families of differing sizes create knapsack-type complementarities at localities. With sizes, a stable matching need not exist, and finding one is computationally in-

tractable. Envy-freeness can always be met but, under multidimensional constraints, it forbids priority violations that harm no one and so generates unnecessary waste. They therefore introduce *interference-freeness*: a family may be envied provided that no higher-priority family that weakly prefers its locality could claim any of the capacity it uses. Interference-freeness is implied by envy-freeness, is logically independent of stability, and is attained in polynomial time by a variant of deferred acceptance, at the cost of tolerating some waste. Our floor axioms follow the same logic. Strict floors, like stability, can fail to exist because of an indivisibility, here a grant’s minimum cost (Section 4.4). Strong feasible floors (Axiom C*) play the role of envy-freeness: they test a shortfall against every eligible grant, including grants far below the merit frontier, so the arrival of a cheap low-merit application can change the allocation (Proposition 10). Feasible floors (Axiom C) play the role of interference-freeness: they test the shortfall only against the claim the priority structure puts next in line, the highest-merit eligible grant not yet fully funded, and the floors-first algorithm attains them in polynomial time (Theorem 2). Two differences are worth noting. Floors are distributional targets rather than priority constraints, and unlike interference-freeness, feasible floors are compatible with non-wastefulness. The closer formal counterpart of interference-freeness in our setting is the first clause of Axiom B, which counts a priority violation only when the funds held by lower-priority grants could actually cover the higher-priority grant. Like these applications, our paper extends matching theory to a setting with complementarities (induced by funding floors), where traditional matroid structures no longer apply.

[Biró, Fleiner, Irving, and Manlove \(2010\)](#) study higher education in Hungary and demonstrate the computational difficulty of matching with upper and lower capacity constraints for schools: they find that determining whether a stable matching exists for this type of problem is NP-complete. Our model may be thought of as having upper and lower capacity constraints, or demands, for applicants, and only an upper capacity constraint for the grant-making institution.

[Kesten and Ünver \(2015\)](#) are the first to study school choice mechanisms that determine lotteries over assignments, which is related to our study of fractional assignment. While student-proposing DA with random tie-breaking is ex-post stable for the school choice problem ([Abdulkadiroğlu & Sönmez, 2003](#)), this paper studies broader ex-ante notions of fairness for stochastic mechanisms. They find a choice rule leading to *strongly ex-ante stable* matching lotteries. That is, they require that a higher priority student gets all the enrollment chance at a school compared to a lower priority student if they desire; moreover, two equal priority students should get an equal probability for a school they would both like, unless one student receives greater probability towards schools they like more. Their first proposed choice rule *fractional deferred acceptance* (FDA) works along the lines of DA, but when equal priority students apply to the same school, if there is not enough capacity, the capacity will

generally be shared as an equal assignment probability between the top-priority students. The second choice rule they study, *fractional deferred acceptance with trading* (FDAT) begins with the FDA outcome, and uses *ex-ante stable improvement cycles* much in the same fashion as the top trading cycle algorithm (Shapley & Scarf, 1974), finding a cycle of students who each prefer another student’s assignment fraction, and trading these fractions accordingly.

Delacrétaz (2021) studies a reserve design context where a social planner has set aside reserves for a number of uses, or categories. Each recipient demands at most one unit, and Delacrétaz proposes drawing that unit evenly from the categories for which the recipient is eligible, a desideratum he calls *category-neutrality*. He shows that the simultaneous reserve allocation is eligibility-compliant, non-wasteful, respects priorities and is category-neutral, and that every allocation with these four properties gives each recipient the same total amount. The characterization is therefore of the aggregate allocation: when categories rank recipients differently, distinct allocations satisfying all four properties can split the same totals differently across categories. The *simultaneous reserve algorithm* builds upon Kesten and Ünver (2015) in assigning fractions of agents iteratively, but it may assign less than one unit to each recipient. In contrast to Kesten and Ünver (2015), which considers many schools but without category-specific reserves, this paper considers multiple reserve categories and one institution; hence it is closer to the grant funding context we consider. In a discrete matching context, the outcome for each agent may be interpreted as the probability of assignment. However, in our context, we take advantage of the fact that budgets are continuous, so a similar rule can be used to determine exact funding amounts assigned from reserve categories to grants. Hence, the section we dedicate to *earmarks* is in the same spirit as Delacrétaz (2021). Our context is less general in that a category’s priority order reflects the overall merit order when comparing grants that are both eligible, as in Sönmez and Yenmez (2019b). This enables a simpler solution: a single pass through the merit order that settles each grant at its own step. In Delacrétaz (2021), the basic *simultaneous reserve algorithm* reallocates capacity across rounds and can cycle without terminating, reaching the simultaneous reserve allocation only in the limit; a variant that solves a linear program between rounds finds it after finitely many rounds, polynomially many in the numbers of recipients and categories. Our model is distinct in that grants are allowed to have arbitrary lower and upper cost bounds, an important detail if grants are meant to be heterogeneous and reflect projects of different scales.

A large literature relaxes strategy-proofness in matching markets, for instance by requiring it only in the limit as the market grows (Azevedo & Budish, 2019; Kojima & Pathak, 2009). Our incentive criteria are of a different kind. Strategy-proofness compares a truthful report with a misreport and evaluates the comparison by the agent’s own preferences; as we argue in Section 4.7, a grant application has

no truthful counterpart, since the applicant authors the project it proposes. We instead compare two applications the same investigator might submit, ordered by the institution’s partial order over types, and require that the rule never fund the worse one while rejecting the better. The closest antecedent is the requirement that a mechanism *respect improvements*, introduced by [Balinski and Sönmez \(1999\)](#) for student placement: a student’s assignment should not deteriorate when her test scores improve. Monotonicity in merit is the analogue of that condition in our setting, and *monotonicity in types* extends it to the dimensions an applicant controls (cost bounds and claimed categories). The condition is also familiar from mechanism design with transfers, where a monotone allocation rule, as in the auction of [Vickrey \(1961\)](#), is what makes truthful reporting implementable; absent transfers, monotonicity is the part of that requirement that remains meaningful. We also consider monotonicity along each dimension separately when the full criterion fails.

Finally, this paper builds on the established insight that incentive requirements place limitations on other desirable properties of a mechanism (Theorem 3). Since this tradeoff is widespread in the literature, we cannot mention all papers that make this observation, and focus on two important examples. First, [Roth \(1982\)](#) finds a tradeoff between stability and incentives for truth-telling for all agents in two-sided matching mechanisms (the marriage problem). This observation has been applied to the school choice context: [Pathak and Sönmez \(2008\)](#) compare the Boston mechanism, which is manipulable, with deferred acceptance, which is strategy-proof. They find that the Boston mechanism rewards sophisticated agents at the expense of sincere ones, so the strategy-proof mechanism may be considered more fair, as it does not disadvantage less sophisticated agents. For problems with floor constraints, we find a tradeoff between a weak incentive criterion (monotonicity in types) and optimality with respect to the social planner’s preferences.

3 Two-Tiered Review

3.1 Department of Defense (DOD)

We consider three main examples of grant-making institutions that each follow the same broad pattern for making funding decisions, (1) review prospective grants to assess their merit, and (2) in light of these assessments and the programmatic categories each application claims, create a balanced choice of funding decisions. First, the Department of Defense funds the Congressionally Directed Medical Research Programs, which gives the following description of their process, “To ensure both scientific excellence and programmatic relevance, the CDMRP administers a two-tiered review process consisting of peer and programmatic review” ([Department of Defense, 2023](#)).

We include an infographic as Figure 1 found on the same webpage, which illustrates their process.



Figure 1: Two-Tiered Review

3.2 National Science Foundation (NSF)

The NSF lists its goals for grant-funding, and provides the following as the first bullet, “Maintaining a balanced portfolio that provides opportunities for original research in all fields of science, engineering, and learning” ([National Science Foundation, 2022](#), p. 31). Like the DOD, the NSF uses the terminology of a “balanced portfolio” as a goal for the second tier of review, which means merit and potential impact alone do not determine funding decisions.

3.3 National Institutes of Health (NIH)

Likewise, the NIH provides clear details about their process, describing the first stage as peer review, and the second level as taking research priorities and portfolio balance considerations¹:

Funding decisions are made through a rigorous dual-level review process. Scientific review groups or study sections, first evaluate and score research proposals for scientific and technical merit. Study sections are made up of volunteer scientists, mostly from academia, and overseen by NIH staff known as scientific review officers.

¹We provide a more detailed description of the NIH’s review and funding policies in Appendix B.

Advisory councils for NIH ICs and the NIH Office of the Director then perform a second-level review for mission relevance. Ultimately, IC directors make final funding decisions, taking into consideration current research priorities and the existing funding portfolio. (Talesnik, 2025)

Additional sources concur with this description of the NIH peer review process (NIH, 2021). Further, this conclusion is supported by the significant mass of funding for grants that fall below the “payline,” meaning many funded grants are lower merit than some rejected grants at that institute (National Institute of Neurological Disorders and Stroke, 2021).

Our model abstracts from the details of these processes and keeps their common structure. The first tier is summarized by a single strict merit order over applications. This is a modeling assumption rather than a feature the sources above establish: review produces scores, which may tie and which some applications never receive, and we take the merit order to be a strict ranking that extends them (see the discussion following Theorem 1). The second tier is summarized by a set of categories, each standing for a programmatic objective, and an application may be eligible for several of them. An application’s categories are those it claims when it is submitted; peer review determines merit but not category membership. Where a category reflects a verifiable status, such as career stage, the claim is simply checked. We consider two ways a policymaker might pursue its portfolio objectives. Under *earmarks*, part of the budget is set aside for each category and may be spent only on grants eligible for it. Under *floors*, each category has a target minimum level of funding for its eligible grants. In both cases, merit enters through two requirements: usable funds should not sit idle, and no grant should be funded ahead of one with a stronger claim to the same funds.

4 Setup

4.1 Earmark Problems

The first type of grant-funding problem we consider is an earmark problem, where funding is sectioned off into reserves whose use is strictly limited to eligible grants.

There is a finite set I of grants applying for funding and a total budget $\mathcal{B} \in \mathbb{R}_+$. The budget is divided among a finite set of categories $\mathcal{C}$, made up of an open category o and the reserved categories $\mathcal{C} \setminus \{o\}$: each category c is given a *reserve* $q_c \in \mathbb{R}_+$, with $\sum_{c \in \mathcal{C}} q_c = \mathcal{B}$. Each grant $i \in I$ has minimal and maximal costs $[b_i, \bar{b}_i] \subseteq (0, \mathcal{B}]$ and is *eligible* for each category in $\rho(i) \subseteq \mathcal{C}$, a set that always contains o . The set $\rho(i)$ is part of the application: the applicant claims the categories its project addresses, and

the institution takes the claim as given (Section 4.7). Grants are ranked by a strict *merit order* $\succ$, and each category c ranks them by a strict *reserve priority order* π_c . The open category ranks by merit alone, $\pi_o \equiv \succ$; a reserved category's order agrees with $\succ$ on the grants eligible for it and ignores the rest. An *earmark problem* collects these components:

$$\mathcal{G} = \langle I, \mathcal{B}, \mathcal{C}, \succ, \rho, \{[\underline{b}_i, \bar{b}_i]\}_{i \in I}, \{q_c\}_{c \in \mathcal{C}}, \{\pi_c\}_{c \in \mathcal{C}} \rangle.$$

A *disaggregate allocation* x is an $|I| \times |\mathcal{C}|$ matrix whose elements $x_{i,c}$ give the assignment from category c 's reserve to grant i , and which satisfies either $x_i = 0$ or $\underline{b}_i \leq x_i \leq \bar{b}_i$, for all grants i , and $0 \leq x_c \leq q_c$, for all categories c . If a matrix x fails one of these conditions, we say x is *unattainable*. We denote the i^{th} row sum $\sum_c x_{i,c}$ as x_i , and the c^{th} column sum $\sum_i x_{i,c}$ as x_c . An *allocation* $\xi \in \mathbb{R}_+^{|I|}$ is a vector of row sums of a disaggregate allocation. An allocation is *attainable* if for all i , either $\xi_i = 0$ or $\underline{b}_i \leq \xi_i \leq \bar{b}_i$. Hence, an allocation gives the total amount of funding assigned to each grant.

A *disaggregate choice rule* $\hat{C}$ maps each earmark problem to a disaggregate allocation, $\hat{C} : \mathcal{G} \mapsto x$, and a *choice rule* C maps each earmark problem to an allocation, $C : \mathcal{G} \mapsto \xi$.

4.2 Floor Problems

A policymaker may want more flexible use of the budget. In a *floor problem*, q_c is no longer a reserve but a *floor*: the approximate minimum amount that should go to grants eligible for category c . Floors need not exhaust the budget, so we drop the open category o and require only $\sum_{c \in \mathcal{C}} q_c \leq \mathcal{B}$; since no category holds funds of its own, we also drop the reserve priority orders π_c . A floor problem collects the remaining components:

$$\mathcal{G} = \langle I, \mathcal{B}, \mathcal{C}, \succ, \rho, \{[\underline{b}_i, \bar{b}_i]\}_{i \in I}, \{q_c\}_{c \in \mathcal{C}} \rangle.$$

Because floors do not assign funds to categories, there is no analogue of a disaggregate allocation. An *allocation* is a vector $\xi \in \mathbb{R}_+^{|I|}$ of funding amounts, and it is *attainable* if it meets the per-grant condition of Section 4.1 and $\sum_{i \in I} \xi_i \leq \mathcal{B}$. For a category c , let $I_c = \{i \in I : c \in \rho(i)\}$ be the grants eligible for c , and let

$$A_c(\xi) = \{i \in I_c : \xi_i < \bar{b}_i\}, \quad d_c(\xi) = q_c - \sum_{i \in I_c} \xi_i$$

be, respectively, the eligible grants that could still absorb funding at ξ and the category's remaining shortfall. A *choice rule* C maps each floor problem to an allocation, $C : \mathcal{G} \mapsto \xi$. A *grant-funding problem* is either an earmark problem or a floor problem.

4.3 Core Axioms

We develop a set of axioms providing the desiderata for each type of problem. We will say a choice rule C satisfies an axiom if it always leads to allocations satisfying the axiom. Axioms A–C will be used to characterize acceptable solutions to floor problems, while Axioms A, B, D and E will be used to uniquely characterize the solution to earmark problems, so Axioms A and B are required for both problems.

Axiom A. *An allocation ξ is **non-wasteful** if there is no attainable $\xi' \neq \xi$ with $\xi' \geq \xi$. In an earmark problem the requirement is placed on the disaggregate allocation, and the comparison ranges only over matrices that respect the designation of funds: there is no attainable $x' \neq x$ with $x' \geq x$ and $x'_{i,c} = 0$ for every grant i and every $c \notin \rho(i)$.*

Non-wastefulness can be checked one grant at a time. Write

$$s(\xi) = \mathcal{B} - \sum_{j \in I} \xi_j \quad \text{and} \quad r_i(x) = \sum_{c \in \rho(i)} (q_c - x_c)$$

for the funds that remain available to grant i : the unspent budget in a floor problem, and the unassigned reserves in i 's own categories in an earmark problem. Then ξ is non-wasteful if and only if, for every $i \in I$,

$$\xi_i = 0 \implies s(\xi) < \underline{b}_i \quad \text{and} \quad \underline{b}_i \leq \xi_i < \bar{b}_i \implies s(\xi) = 0, \quad (1)$$

and a disaggregate allocation x with $x_{i,c} = 0$ for every $c \notin \rho(i)$ is non-wasteful if and only if (1) holds with x_i and $r_i(x)$ in place of ξ_i and $s(\xi)$.² That is, an unfunded grant is passed over only if what remains available to it is below $\underline{b}_i$, and a funded grant stops short of $\bar{b}_i$ only if nothing remains.

The earmark clause compares x only with matrices that respect designation, since an expansion drawing on a category a grant is not eligible for is not one the institution may carry out. Axiom B below counts available funds the same way, through $r_i(x)$. This is the *constrained* form of non-wastefulness, as in Ehlers et al. (2014).

Combining the merit total order with categories helps determine which grants have priority for funding. For earmark problems, we already defined the reserve priority order, defining which grants have higher priority for each reserve. On the other hand, floor problems do not subdivide the budget into sections clearly designated to each category. Therefore, in floor problems there is a priority partial order $\succ_\rho$, where a grant has higher priority than another grant if it is eligible for a superset of the categories and has strictly higher merit.

²If an attainable $x' \geq x$ differs from x , pick a grant i with $x'_i > x_i$ and let x'' agree with x' in row i and with x elsewhere. Every column sum of x'' is at most that of x' , so x'' is attainable, respects designation, and expands x in row i alone. Such an expansion exists exactly when $r_i(x) \geq \underline{b}_i$ if $x_i = 0$, and when $r_i(x) > 0$ if $\underline{b}_i \leq x_i < \bar{b}_i$. The floor case is identical with $s(\xi)$ in place of $r_i(x)$.

Axiom B. In a floor problem, an allocation ξ *respects priorities* if

1. for every $i \in I$ with $\xi_i = 0$, there is no $J \subseteq \{j \in I : \xi_j > 0 \text{ and } i \succ_\rho j\}$ with $\sum_{j \in J} \xi_j \geq \underline{b}_i$; and
2. for every $i \in I$ with $\underline{b}_i \leq \xi_i < \bar{b}_i$, there is no $j \in I$ with $i \succ_\rho j$ and $\xi_j > 0$.

In an earmark problem the same requirement is imposed reserve by reserve: a disaggregate allocation x respects priorities if

1. for every $i \in I$ with $x_i = 0$, there is no $J \subseteq I$ containing some j with $x_{j,c} > 0$ for a category c having $i \pi_c j$, such that $\sum_{j \in J} \sum_{c: i \pi_c j} x_{j,c} + \sum_{c \in \rho(i)} (q_c - x_c) \geq \underline{b}_i$; and
2. for every $i \in I$ with $\underline{b}_i \leq x_i < \bar{b}_i$, there is no $j \in I$ with $i \succ j$ and $\sum_{c \in \rho(i)} x_{j,c} > 0$.

Each clause forbids one kind of priority violation. The first says that no collection of funds going to grants with a worse claim—together, in an earmark problem, with the reserves still standing in i 's own categories—could have covered i instead. The second says that once a grant with a better claim has been funded but not to its ceiling, nothing should go to a grant with a worse claim. This is the counterpart of “eliminates justified envy” in the matching literature, with the qualification that merit alone does not settle who has the better claim: a grant of higher merit that is eligible for fewer categories is not comparable under $\succ_\rho$, and neither grant has a claim against the other.

The earmark version of the first clause carries a proviso: J must contain some grant j holding funds from a category c with $i \pi_c j$. Without it, $J = \emptyset$ would be admissible, and the clause would rule out every allocation in which the unassigned reserves in i 's own categories, $\sum_{c \in \rho(i)} (q_c - x_c)$, are by themselves at least $\underline{b}_i$. Axiom A already rules these out, since i could be funded from those reserves without taking anything from anyone, and no grant with a worse claim is involved. The proviso ensures that every violation of the clause involves a grant with a worse claim holding funds i could have used; once such a grant is present, unassigned reserves in i 's categories still count toward covering i . The floor version needs no proviso: its sum contains no unassigned budget, so an empty J contributes $0 < \underline{b}_i$.

Non-wastefulness and respecting priorities are the universal baseline for both earmark and floor problems. Any acceptable allocation should make full use of the funds at hand, and it should respect the priority order.

4.4 Floor-Specific Axioms

While quotas, upper bounds on funding, may arguably aid in the effort of creating a balanced portfolio of grants, this is unlikely to be the case when grants are eligible

for multiple categories. For instance, if a grant i is eligible for categories c and c' , it would be undesirable to withhold funding towards i due to reaching a quota in c while there is slackness for the floor in c' . Hence, we focus on floors alone.

There is one caveat to implementing floors. Here, there are minimum costs towards each grant. For example, there may be three grants costing 4 each, and a budget of 10. If the first two grants are eligible for category c_A and the last is eligible for category c_B , and floors are 5 for each of c_A and c_B , there is no way to meet floors; either 8 goes towards category c_A and zero towards c_B , or 4 goes to each. As a result, it is important to develop an acceptable alternative formulation of floors, which must be *feasible* floors in that they are not strict, but may always be met.

Axiom C. *An allocation ξ **satisfies feasible floors** if for every category $c \in \mathcal{C}$ with $d_c(\xi) > 0$, the highest-merit $i^* \in A_c(\xi)$ satisfies $\xi_{i^*} = 0$ and $\underline{b}_{i^*} > d_c(\xi)$.*

Axiom C*. *An allocation ξ **satisfies strong feasible floors** if for every category $c \in \mathcal{C}$ with $d_c(\xi) > 0$ and every $i \in A_c(\xi)$, $\xi_i = 0$ and $\underline{b}_i > d_c(\xi)$.*

Axiom C* requires that whenever category c falls short of its floor, every eligible grant that could still absorb funding is unfunded and has a minimum cost exceeding the shortfall $d_c(\xi)$, so that funding any of them would carry c past its floor. Axiom C imposes this requirement only on the highest-merit grant in $A_c(\xi)$, the grant the institution would fund next in category c , and is therefore a relaxation of Axiom C*. If every grant eligible for c is fully funded, then $A_c(\xi) = \emptyset$ and both axioms hold vacuously for c . We impose Axiom C throughout. The stronger axiom is incompatible, given Axioms A and B, with the independence property introduced in Section 4.5 (Proposition 10, proved in Section A.4).

For floor problems, we will find a choice rule that is non-wasteful, respects priorities, and satisfies feasible floors. However, these axioms are insufficient for a unique characterization. We will achieve a unique solution by making use of a total order of allocations induced by the merit of grants in Section 7.

4.5 Independence of Non-contending Grants

Fix a floor problem $\mathcal{G}$, let $\xi = C(\mathcal{G})$, and let j satisfy $\xi_j = 0$ (this includes any $j \notin I$). Then j is *non-marginal at ξ* if

1. $\underline{b}_j > \mathcal{B} - \sum_{i \in I} \xi_i$, and
2. for every $c \in \rho(j)$ there is some $i \in I_c$ with $i \succ j$ and $\xi_i < \bar{b}_i$.

Otherwise j is *marginal at ξ* . Funded grants are not classified either way: the distinction concerns which unfunded grant the institution would turn to next, so it has no content for a grant already receiving funds.

Say that j is *non-contending at ξ* if it is non-marginal at ξ and, in addition, $i \succ j$ for every $i \in I$ with $\xi_i > 0$, so that j falls below the funded frontier. We state the third requirement separately from the first two because the two notions do different work: non-marginality alone drives monotonicity in types (Proposition 5), while the property defined below requires a grant to be out of contention altogether.

If $j \notin I$, write $\mathcal{G}^{+j}$ for the problem obtained from $\mathcal{G}$ by adjoining j to I . A choice rule C is *independent of non-contending grants* if, for every floor problem $\mathcal{G}$ and every application $j \notin I$ that is non-contending at $C(\mathcal{G})$, $C(\mathcal{G}^{+j}) = (C(\mathcal{G}), 0)$.

In any allocation the institution has exactly two kinds of margin. There is the budget margin, funds sitting unassigned, and there is the merit frontier: the best grant that has not yet been fully funded, which may be considered for each category and overall. Every funding decision the institution can make is a decision at one of these margins: spend idle money, or move down the queue for some category or overall. A non-marginal application sits at neither. It is too large to absorb the idle funds, and in every eligible category there is a better grant with higher priority seeking funding. A non-contending application satisfies one further requirement: it ranks below every grant that actually received funding, so it could not have displaced any of them. Such an application was never in contention, and its addition should not affect funding decisions.

Axiom C*, strong feasible floors, strengthens Axiom C by quantifying over every eligible grant that could still absorb funding rather than the highest-merit one alone; since the binding case is the cheapest such grant, more funding is required before a floor counts as met. In Section A.4 we find that if a choice rule is non-wasteful, respects priorities, and satisfies strong feasible floors, it fails to be independent of non-contending grants (Proposition 10). The reason is that a threshold set by the cheapest eligible application rather than the best one is disturbed by the arrival of a cheap low-merit grant: a floor previously deemed satisfied now counts as unmet, and closing the gap requires diverting funding into that category and away from higher-merit grants. This represents a normative failure since the addition of low-quality grants alters the allocation. Furthermore, it drastically complicates the problem of finding a suitable choice rule since there is no longer a clear margin for decision-making.

4.6 Earmark-Specific Axioms

Axiom D. A disaggregate allocation x is **eligibility-compliant** if $x_{i,c} = 0$ whenever $c \notin \rho(i)$.

Funding is not used in a way that violates its designation: a grant draws only on the reserves of categories it is eligible for. Since reserve categories have a priority order that prefers to go unassigned over funding a grant outside the category, this axiom

agrees with the typical notion of *individual rationality*. Funding is not explicitly split into categories in a floor problem, so the axiom is relevant only to earmark problems.

Axioms A and D are therefore meant to be read together. Eligibility-compliance restricts which reserves may fund a grant. Non-wastefulness compares x only with eligibility-compliant matrices, so it rules out only those expansions that draw on reserves the grants are eligible for. Without that restriction, Axiom A could require expansions that Axiom D forbids, and on some problems no allocation would satisfy both. This happens, for example, when a grant funded below its ceiling could be topped up from an unassigned reserve it is not eligible for, and no eligibility-compliant allocation can use that reserve up.³

Axioms A and B then divide the work without overlapping. Axiom A governs idle designated funds: no grant may absorb more from the reserves of the categories it is eligible for, whether it is unfunded or funded short of its ceiling. Axiom B governs funds already committed: nothing may be held by a worse claim while a better one goes unmet, and idle reserve enters its first clause only as a supplement to assigned funds.

Axiom E. Writing $\hat{I}_{i,c}$ for the grants of higher merit than i that are eligible for c , a disaggregate allocation x is **category-neutral** if for every grant i and every category $c \in \rho(i)$,

$$x_{i,c} < \max_{c' \in \rho(i)} x_{i,c'} \quad \text{only if} \quad x_{i,c} + \sum_{i' \in \hat{I}_{i,c}} x_{i',c} = q_c.$$

Category-neutrality, which follows [Delacrétaz \(2021\)](#), asks the categories a grant is eligible for to share its cost evenly where they can. The one exception is a category that has run out: c may contribute less to i than another eligible category only if c 's reserve is fully allocated to i and the grants of higher merit that are eligible for c . For earmark problems, we will show that there is a unique choice rule that is non-wasteful, respects priorities, and is eligibility-compliant and category-neutral.

4.7 Type Monotonicity

We treat a grant's funding request and claimed categories as choices of the applicant rather than facts the institution could in principle verify. Its merit score is different: review assigns it, and the applicant influences it only through the project it proposes, a distinction we return to below. Treating the request and the categories as choices is

³For a minimal instance, let $I = \{i_1\}$ and $\mathcal{C} = \{o, c_A\}$ with $q_o = 3$ and $q_{c_A} = 5$, so $\mathcal{B} = 8$, and let $\rho(i_1) = \{o\}$ with $[\underline{b}_{i_1}, \bar{b}_{i_1}] = [1, 8]$. The best eligibility-compliant allocation gives $x_{i_1,o} = 3$. Yet the matrix with $x_{i_1,o} = 3$ and $x_{i_1,c_A} = 5$ has row sum $8 \in [1, 8]$ and exceeds no reserve, so under an unrestricted comparison that allocation would count as wasteful—as would every other eligibility-compliant one. The restricted comparison is in any case the notion the proofs in Appendix A verify.

a modeling stance, not a property the institutional descriptions in Section 3 establish, and the argument for it is the following. An applicant does not discover the cost of its project and report it; it decides how large a project to propose, over what range of funding it is willing to carry the work out, and which objectives to claim relevance to. There is accordingly no truthful report against which a submission may be compared, and incentive compatibility in the revelation sense is ill-posed here rather than merely unattainable. The extreme case makes the point: a lone applicant eligible for all categories facing a budget $\mathcal{B}$ may propose a project of any size up to $\mathcal{B}$, and any non-wasteful rule funds it; likewise the highest-priority grant may always request more and, under a rule that respects priorities, receive it. Nothing distinguishes an inflated request from a genuinely larger project, because the applicant authors both. Nor are there transfers with which the institution might make an applicant bear the cost of the scale it chooses.

What remains is a question about design rather than disclosure. Does the mechanism reward applicants for proposing larger, costlier, or less flexible projects than the work warrants? We approach this comparatively: our criteria rank two applications the same investigator might submit and require that the rule not favor the one that is worse from the institution's standpoint.

An application presents the institution with a *type* $\theta = (\underline{b}, \bar{b}, R, m)$: a funding request $[\underline{b}, \bar{b}]$, a set of claimed categories $R \subseteq \mathcal{C}$, and the merit score m returned by peer review. The applicant chooses the first two, and the third follows from the project it proposes. Grant i 's type is $\theta_i = (\underline{b}_i, \bar{b}_i, \rho(i), m_i)$, and, keeping with convention, θ_{-i} denotes the types of all other grants. The set of merit scores $\{m_i\}_{i \in I}$ determines the merit order: $i \succ j$ if and only if $m_i > m_j$, and we assume $m_i \neq m_j$ whenever $i \neq j$. This is the strictness assumption discussed after Theorem 1; actual review scores can tie, and m_i should then be read as a score that also encodes the institution's tie-breaking. Holding the rest of $\mathcal{G}$ fixed, we write $C(\theta, \theta_{-i})$ for the allocation when i submits θ , and say that C funds i at θ if $C(\theta, \theta_{-i})_i > 0$.

For types $\theta = (\underline{b}, \bar{b}, R, m)$ and $\theta' = (\underline{b}', \bar{b}', R', m')$, write $\theta \preceq_t \theta'$ if $\underline{b} \leq \underline{b}'$, $\bar{b} \leq \bar{b}'$, $R \supseteq R'$, and $m \geq m'$. This partial order captures when one application is weakly "better" than another. (Notice, this is different from the priority partial order for floor problems $\succ_\rho$, which does not consider cost.)

A choice rule C is *monotonic in types* if for every grant i , every θ_{-i} , and every $\theta \preceq_t \theta'$, C funds i at θ whenever it funds i at θ' . This guarantees that improvements in a grant lead to better outcomes in terms of receiving funding or not. We may also consider monotonicity with respect to each component of the type: C is *monotonic in cost*, *in merit*, or *in categories* if the same holds for all pairs $\theta \preceq_t \theta'$ that differ only in $(\underline{b}, \bar{b})$, only in m , or only in R , respectively. Finally, C is *monotonic in types for non-marginal grants* if the same holds whenever i is non-marginal at $C(\theta, \theta_{-i})$ (Section 4.5).

These conditions do not reach the applicant in the same way. The funding request and the claimed categories are written into the application itself, so monotonicity in cost and in categories concerns how an applicant presents a given project. Merit is not written in: review assigns it, and the applicant affects it only through the project it proposes and, via net value, the scale it requests. Monotonicity in merit therefore asks that the rule not overturn the judgment of review, never funding an application it would reject had review thought better of it, so that no applicant gains by proposing weaker work.

What merit measures

It is worth fixing ideas about what merit measures. Each grant is endowed with some gross value at its requested funding level, $V_i(\bar{b}_i)$. At higher requested levels a project is able to take on more scope and is more likely to provide social value. For instance, more lab materials means more trials are allowed, or more hours of labor means a survey sample may be expanded. It follows that V_i is increasing in $\bar{b}_i$. On the other hand, each dollar assigned to fund project i has an opportunity cost, which we assume is constant κ . Peer review should therefore rank the merit of grants based upon their net value,

$$m_i = V_i(\bar{b}_i) - \kappa \cdot \bar{b}_i,$$

and $i \succ j$ whenever $m_i > m_j$. Nothing below depends on the form of V_i : the order $\succ$ that review induces is a primitive.

Note that V_i may rise at rate κ over a range of requested levels, in which case m_i is constant there and an applicant may ask for more while its merit is unchanged.

Merit does not depend on $\underline{b}_i$. Review evaluates the project at the level requested; the minimum operable scale describes a contingency that review never assesses, and it does not enter the net value above. It matters only at the selection stage, through attainability, through non-wastefulness, and through the threshold at which a category's floor is deemed satisfied (Axiom C).

Taxonomy of deviations

An applicant may submit a different application in several ways, and our criteria do not treat them alike. Write θ_i for the original application and θ'_i for the alternative.

Type	Alternative submitted	$\underline{b}_i$	b_i	m_i	vs. θ_i
<i>i</i>	Larger request, lower merit	—	↑	↓	disimprovement
<i>ii</i>	Larger request, same merit	—	↑	—	disimprovement
<i>iii</i>	Larger request, greater merit	—	↑	↑	incomparable
<i>iv</i>	Smaller request, lower merit	—	↓	↓	incomparable
<i>v</i>	Larger minimum, same merit	↑	—	—	disimprovement

Table 1: Deviation Taxonomy

We call a deviation of types *i–iii* a *moonshot*, characterized by an increase in the scale of the project, and hence an increase in the gross value. Types *i* and *ii* are *inefficient moonshots*, where the larger scale leads to weakly lower merit; while the gross value rises, it is matched or outpaced by the opportunity cost κ . In the knife-edge case of type *ii*, the gross value rises at the rate κ , so merit is unchanged by an expansion in the request. It is undesirable to incentivize expansions that keep merit equal because it makes the planner’s problem harder; when grants are large relative to the budget, the tradeoff to funding that grant is larger. For instance, if a grant is eligible for only a strict subset of the categories, it may be difficult to justify committing an outsized share of the budget to that project; it could pose a challenge to meet the other category floors, or the project could eat up all eligible earmarks plus the open category earmark. Therefore, type *ii* is a disimprovement in type. Type *iii* is an *efficient moonshot*, where the larger scale is justified by improved merit. It is ambiguous whether this is an improvement; the grant is more costly, so may be more difficult to fit into the budget, but has better bang for its buck. In a vacuum, this would be an improvement, but when fitting into a balanced portfolio, it is unclear.

Type *iv* is a smaller request that review scores lower, so it is incomparable. Perhaps it is easier to fit the grant into the budget, but the project is less worthwhile. Finally, type *v* is an unambiguous disimprovement: the grant has the same ceiling and the same merit, but a larger minimum. The application is less flexible about the scale at which the work can be carried out, which leaves the institution fewer ways to fit it into the budget, with no gain in merit to offset it. A choice rule should not incentivize this behavior.

Notice that two of the pathologies (types *ii* and *v*) exhibit a deviation only in cost; therefore, monotonicity in cost will prevent the pathology from leading to a better outcome. On the other hand, a deviation of type *i* cannot produce a better outcome under a rule which is monotonic in types.

One further limitation is worth stating. Our monotonicity conditions concern only whether C funds a grant, not how much it receives. An application funded under both proposals, but funded more generously under the worse one, is invisible to them. Type *v* is where this matters most: a grant cannot secure funding by raising $\underline{b}_i$ under a rule

monotonic in types, but it may enlarge an award it would have received in any case.

Each of these conditions concerns a grant's own application. A separate requirement, independence of non-contending grants (Section 4.5), concerns the pool rather than the submission: whether an application that was never in contention can change the allocation merely by being submitted. The two are taken up together when we compare choice rules in Section 7.

5 Earmarks

When funding is blocked off into sections for each category, but grants may be eligible for multiple categories, how should funding be afforded to grants? When category-neutrality is imposed, requiring that eligible categories evenly share the cost of grants when possible, there is only one solution, utilizing the category-neutrality axiom from Delacrétaz (2021).

We study an earmark problem, which takes the form

$$\mathcal{G} = \langle I, \mathcal{B}, \mathcal{C}, \succ, \rho, \{[\underline{b}_i, \bar{b}_i]\}_{i \in I}, \{q_c\}_{c \in \mathcal{C}}, \{\pi_c\}_{c \in \mathcal{C}} \rangle,$$

where I is a set of grants, $\mathcal{B}$ is the total budget, $\mathcal{C}$ is the set of categories, ρ is the category-eligibility function, $[\underline{b}_i, \bar{b}_i]$ are the grant costs, q_c gives the earmark for category c , and π_c gives category c 's priority order. We will find a rule assigning these funds to grants, where the assignment of each pool q_c respects the priority order π_c .

To describe the disaggregate choice rule satisfying Axioms A, B, D and E in words, it works by ordering the grants according to merit, then starting with the highest-merit grant, checking whether it is affordable via reserves from its categories. If it is affordable, fund it as much as possible, and in a way that evenly shares costs, so the allocation is *category-neutral*; repeat this step until all grants have been considered. We illustrate the solution procedure with an example before its rigorous statement.

Example 1. Suppose that the total budget is 10 and $q_{c_1} = q_{c_2} = 4$, so $q_o = 2$. Assume grants $i_1 \succ i_2 \succ i_3 \succ i_4$ have lower and upper costs and categories as follows:

	i_1	i_2	i_3	i_4
$[\underline{b}_i, \bar{b}_i]$	$[1/2, 1]$	$[1, 3]$	$[4, 5]$	$[3, 4]$
$\rho(i)$	$\{o, c_1\}$	$\{o\}$	$\{o, c_1\}$	$\{o, c_1, c_2\}$

Then the solution procedure works as follows:

	Step 1			Step 2		
	$o(2)$	$c_1(4)$	$c_2(4)$	$o(2)$	$c_1(4)$	$c_2(4)$
i_1	1/2	1/2	0	1/2	1/2	0
i_2	0	0	0	3/2	0	0
i_3	0	0	0	0	0	0
i_4	0	0	0	0	0	0

	Step 3			Step 4		
	$o(2)$	$c_1(4)$	$c_2(4)$	$o(2)$	$c_1(4)$	$c_2(4)$
i_1	1/2	1/2	0	1/2	1/2	0
i_2	3/2	0	0	3/2	0	0
i_3	0	0	0	0	0	0
i_4	0	0	0	0	2	2

In this example, and in any, this solution procedure should be straightforward because it is an immediate acceptance rule, where the only change during step k is in grant i_k , which receives funding (or is skipped, as with i_3 in the example) solely based on the available reserves in its categories. We define the *serial-SR disaggregate choice rule* C^{SR} as the mapping from earmark problems to outcomes x of the serial-simultaneous reserve (SR) algorithm.

Assume $i_k \succ i_{k+1}$, for all $k < n = |I|$. Let $x^0 = 0 \in \mathbb{R}^{n \times |\mathcal{C}|}$ be the starting matrix of funding, which will be updated one row at a time until $t = n$, in which case let x denote the final result.

Step $t \geq 1$:

For $i \neq i_t$ and for all c , let $x_{i,c}^t = x_{i,c}^{t-1}$. If $\sum_{c \in \rho(i_t)} (q_c - x_c^{t-1}) < \bar{b}_{i_t}$, set $x_{i_t,c}^t = 0$ for all c . Otherwise, set $x_{i_t,c}^t = 0$ for $c \notin \rho(i_t)$, and choose $\{x_{i_t,c}^t\}_{c \in \rho(i_t)}$ to solve

$$x_{i_t}^t = \min\{\bar{b}_{i_t}, \sum_{c \in \rho(i_t)} (q_c - x_c^{t-1})\} \quad (2)$$

$$x_{i_t,c}^t = \min\{q_c - x_c^{t-1}, \max_{c' \in \rho(i_t)} \{x_{i_t,c'}^t\}\}. \quad (3)$$

Algorithm 1: serial-SR algorithm

Theorem 1. *A disaggregate choice rule satisfies Axioms A, B, D and E (non-wastefulness, respecting priorities, eligibility-compliance, and category-neutrality) if and only if it is the serial-SR choice rule C^{SR} .*

Two features of Theorem 1 go beyond [Delacrétaz \(2021\)](#). The first is uniqueness of the disaggregate allocation. Delacrétaz characterizes the simultaneous reserve outcome at the level of the aggregate allocation: every allocation satisfying his four axioms gives each recipient the same total, but when two categories rank recipients in opposite orders the axioms admit a continuum of allocations that split those totals differently between the categories. Here the four axioms pin down the whole matrix x , so they determine not only how much each grant receives but also how its funding is charged to categories. The second is a single-pass procedure. Because every category’s priority order is the restriction of one merit order $\succ$, the serial-SR algorithm settles each grant once, at its own step, and halts after $|I|$ of them, each step an explicit computation from the residual reserves (Proposition 4). In [Delacrétaz \(2021\)](#), where categories may rank recipients independently, the basic simultaneous reserve algorithm can cycle without terminating and reaches the allocation only in the limit; finite termination there is obtained by a variant that solves a linear program between rounds. The common merit order is what buys both, and it is a maintained modeling assumption: we take review to deliver a strict ranking of all applications, and a category’s priority order to be that ranking restricted to the grants eligible for it. The institutional descriptions in Section 3 and Appendix B establish a first stage of review that scores applications; they do not by themselves deliver a single strict ranking. At the NIH, for instance, two discussed applications may receive the same overall impact score, and applications that are not discussed receive no score at all. Our results hold for any fixed strict order, so in such an application the order should be read as a strict ranking that extends review’s scores to every application. How the institution breaks ties, and where it places unscored applications, is a choice outside the model, and different choices can produce different allocations. The same descriptions also do not establish that every category ranks its eligible applications identically: the second tier weighs programmatic priorities, and we model those priorities through the categories and their reserves rather than through category-specific reorderings of merit. Were categories to rank applications differently, the problem would return to the more general setting of [Delacrétaz \(2021\)](#): the axioms would no longer pin down the disaggregate allocation, and a single pass through the grants would no longer suffice to compute it.

A third feature is that the four axioms are a minimal set for the characterization: drop any one of them and the remaining three admit an allocation they cannot exclude. This is worth stating because Axioms A and D are close neighbors—Axiom A quantifies over eligibility-compliant matrices, so it is stated in terms that Axiom D supplies—and one might suspect that the pair says one thing twice. It does not, in either direction. An allocation may draw only on categories a grant is entitled to and still leave designated reserves idle, and an allocation may spend a reserve on an ineligible grant while leaving nothing idle at all.

Proposition 1. *None of Axioms A, B, D and E is implied by the other three. For each of them there is an earmark problem and an attainable disaggregate allocation that satisfies the other three axioms and fails that one.*

The fact that the axioms are satisfied should be intuitive. Eligibility-compliance is enforced by construction, and non-wastefulness is implied because each grant i receives as much funding as the reserves of its eligible categories permit, unless too little remains in those categories to meet the lower cost bound $\underline{b}_i$. Since Axiom A compares x only against eligibility-compliant matrices, the reserves standing in categories outside $\rho(i)$ are not funds i was passed over for. Moreover, in each step, if we have already achieved the maximal degree of funding towards higher-merit grants, then there is only one appropriate amount to give to the next best grant without violating Axiom B, and category-neutrality dictates how the cost will be split between eligible categories. Furthermore, if a grant was passed up on its turn due to being too expensive, it will never be affordable in a future step; this allows consideration of past funding decisions during some step t to be equivalent to all final funding decisions towards higher-merit grants than i_t .

Category-neutrality is not free, however. Axiom A tests cellwise expansions of x —whether some grant could draw more from a reserve it is eligible for, with every other entry held fixed—and not reallocations of the charges already placed on the categories. When grants have positive minimum funding requirements, Axiom E can leave usable reserves idle that a different eligible split of the same charges would put to use. Example 1 already shows this. Serial-SR spends $1 + 3/2 + 4 = 13/2$ of the budget of 10 and leaves $7/2$ unspent: $3/2$ in c_1 and 2 in c_2 . Yet the eligibility-compliant allocation that funds i_1 and i_2 at one unit each from o , i_3 at 4 from c_1 , and i_4 at 4 from c_2 funds all four grants, exhausts the budget, and satisfies Axioms A, B and D. It fails only Axiom E, because i_1 is charged entirely to o while c_1 has reserve to spare. A smaller instance isolates the mechanism.

Example 2. *Let $\mathcal{C} = \{o, c_A\}$ with $q_o = q_{c_A} = 1$, so $\mathcal{B} = 2$, and let $i_1 \succ i_2$ with $[\underline{b}_{i_1}, \bar{b}_{i_1}] = [\underline{b}_{i_2}, \bar{b}_{i_2}] = [1, 1]$, $\rho(i_1) = \{o, c_A\}$ and $\rho(i_2) = \{o\}$. Serial-SR charges i_1 half to each category, $x_{i_1,o} = x_{i_1,c_A} = 1/2$. Grant i_2 then faces $1/2 < \underline{b}_{i_2}$ in o and goes unfunded, and half the budget is left idle. Charging i_1 entirely to c_A instead leaves all of o for i_2 , funds both grants, and exhausts the budget.*

Neither outcome violates Axiom A as defined: at C^{SR} the half unit left in o cannot be given to i_2 , whose minimum it does not meet, nor to i_1 , which is at its ceiling. The loss is therefore not a failure of non-wastefulness but an efficiency cost of Axiom E itself, and it comes from minimum funding requirements. If $\underline{b}_{i_2} \leq 1/2$, the half unit left in o would go to i_2 , as it would in the fractional setting of [Delacrétaz \(2021\)](#), where a recipient can absorb any fraction of a unit. With lumpy grants, splitting

the charge of a grant eligible for several categories evenly can leave a residual in a category that is too small for the grants that still need it. Avoiding this requires charging such a grant to the categories that lower-merit applicants cannot use, so that how a grant's funding is split would depend on applications ranked below it, and the resulting rule is not category-neutral; by Proposition 1, Axioms A, B and D alone do not single out an allocation. Category-neutrality should accordingly be read as a substantive choice. It treats the categories symmetrically in how they bear the cost of shared grants, so that no category subsidizes another, and the price is that some designated funds may go unspent that a different apportionment of the same grants would use.

The rule is also well behaved with respect to the applications it receives. Establishing this calls for a different kind of argument, one that compares the allocations produced by two problems differing only in the report of a single grant.

Proposition 2. *The serial-SR choice rule C^{SR} is monotonic in types.*

In other words, the disaggregate choice rule never penalizes a grant for a weak improvement: no deviation that Table 1 classifies as a disimprovement—types i , ii and v —can secure funding for a grant that would otherwise go unfunded.

Because funding decisions for a grant are determined only by funding towards higher-merit grants, C^{SR} satisfies a consistency property analogous to independence of non-contending grants, and in fact a stronger one. In an earmark problem the funds available to j are the residual reserves of its own categories rather than the residual budget, so the affordability condition takes the form $\underline{b}_j > \sum_{c \in \rho(j)} (q_c - x_c)$; no analogue of the second condition of non-marginality is required.

Proposition 3. *Let $\mathcal{G}$ be an earmark problem, let $x = C^{\text{SR}}(\mathcal{G})$, and let $j \notin I$ satisfy $i \succ j$ for every $i \in I$ with $x_i > 0$. Then $C^{\text{SR}}(\mathcal{G}^{+j})_i = x_i$ for every $i \in I$. If in addition $\underline{b}_j > \sum_{c \in \rho(j)} (q_c - x_c)$, then $C^{\text{SR}}(\mathcal{G}^{+j}) = (x, 0)$.*

The first part says the addition of any grant ranked below every funded grant leaves the incumbent allocation untouched; the second says such a grant is itself unfunded whenever it cannot be covered by the reserves still standing in its categories. Together they give the earmark counterpart of independence of non-contending grants. Note that the first part holds without any affordability requirement at all, which is the sense in which C^{SR} satisfies more than the floor-problem property asks.

This concludes the discussion of earmarks. We have illustrated how fair and non-wasteful allocation of earmarked funds is uniquely possible if one desires that eligible reserve categories share the cost evenly, although, as Example 2 shows, even sharing can leave usable reserves idle when grants have minimum funding requirements. Further, Proposition 2 and Proposition 3 say this unique solution does not lead to

glaring incentive problems: it never funds a weaker application it would otherwise reject, and it is unaffected by applications that were never in contention. In contrast, studying problems with floors will not lead to such a neat characterization.

5.1 Computation

The serial-SR algorithm considers each grant exactly once, in merit order, and never revisits a funding decision. All the work within a step consists of splitting one grant's funding among its eligible categories, and the only non-trivial operation that requires is sorting the categories by how much reserve they have left.

Proposition 4. *The serial-SR algorithm operates in polynomial time with a total time complexity of $O(|I| \cdot |\mathcal{C}| \cdot \log(|\mathcal{C}|))$ and terminates in $|I|$ steps.*

6 Floors

A floor problem takes the form

$$\mathcal{G} = \langle I, \mathcal{B}, \mathcal{C}, \succ, \rho, \{[b_i, \bar{b}_i]\}_{i \in I}, \{q_c\}_{c \in \mathcal{C}} \rangle,$$

where I is a set of grants, $\mathcal{B}$ is the total budget, $\mathcal{C}$ is the set of categories, ρ is the category-eligibility function, $[b_i, \bar{b}_i]$ are the grant costs, and q_c gives the floor for each category, an approximate minimum guarantee. We focus on *choice rules* which map a grant-funding problem $\mathcal{G}$ to an allocation ξ . We illustrate how to find an allocation that is non-wasteful, respects priorities, and satisfies feasible floors.

6.1 Algorithmic Construction

A planner who wants to meet every floor, in the sense of feasible floors, while funding the best grants has two natural options, illustrated in Figure 2. In the *sequential* procedure (case 1), the planner takes the categories in a fixed order and, for each in turn, funds its highest-merit grants until its floor is met. In the *simultaneous* procedure (case 2), the planner repeatedly funds the highest-merit grant eligible for any category whose floor is not yet met, updating which floors are met after each grant. In both, the budget left once every floor is met (the crosshatched block) goes to the highest-merit remaining grants regardless of category.

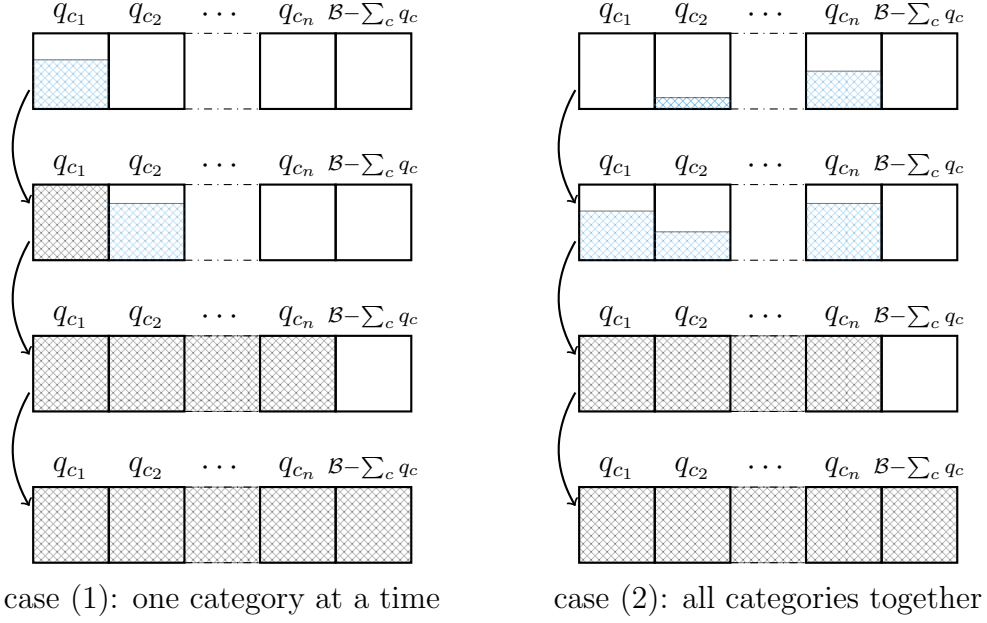
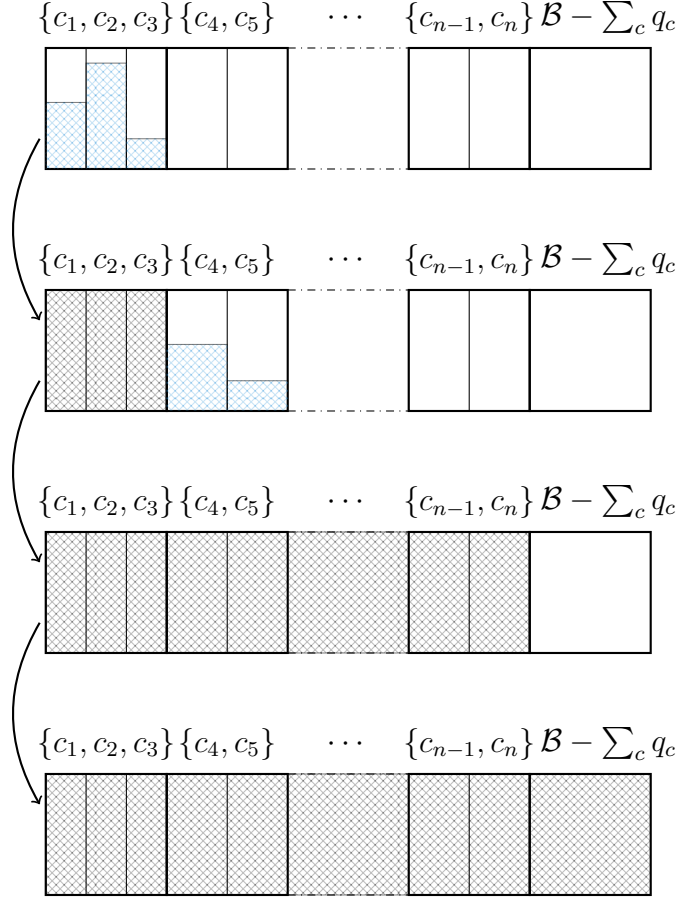


Figure 2: *Two procedures for meeting floors.* Each box is a category's floor q_c ; the crosshatched block is the remaining budget $\mathcal{B} - \sum_c q_c$. Arrows show the order in which funds are assigned.

We will show that either procedure yields a non-wasteful allocation that respects priorities and satisfies feasible floors. Both are special cases of a single procedure. Partition the categories into ordered groups, for example $\{c_1, c_2, c_3\}, \{c_4, c_5\}, \dots, \{c_{n-1}, c_n\}$, and apply the simultaneous procedure to one group at a time (Figure 3); the remaining budget then goes to the highest-merit remaining grants regardless of category.



General case: one group of categories at a time

Figure 3: *General procedure.* Categories are partitioned into ordered groups, and the simultaneous procedure is applied to one group at a time before the remaining budget (crosshatched) is assigned.

Groups of size one give the sequential procedure, and a single group gives the simultaneous one. We work with this general procedure for the rest of the section. To find an allocation satisfying the axioms, we will iteratively build a portfolio of chosen grants as described, so the notion of which grants may be added is useful. A grant i is in the ξ -**budget set** $\mathcal{B}_\xi$ if either $[\xi_i = 0 \text{ and } \sum_{j \in I} \xi_j + \underline{b}_i \leq \mathcal{B}]$, or $[\underline{b}_i \leq \xi_i < \bar{b}_i \text{ and } \sum_{j \in I} \xi_j < \mathcal{B}]$. In other words, i belongs to the ξ -budget set if it is possible to add funds to i when ξ is the current assignment; equivalently, i does not belong to $\mathcal{B}_\xi$ if $\xi_i = 0$ and $\underline{b}_i$ exceeds the amount left unassigned, or i has received some funding less than its upper bound $\bar{b}_i$, but the entire budget is already assigned

in ξ , or i is already fully funded, $\xi_i = \bar{b}_i$.

6.2 Floors-First Choice Rule

Suppose there is an ordered partition $\Delta = \{\delta^1, \dots, \delta^N\}$ of the categories $\mathcal{C}$, where the δ are disjoint and have $\bigcup_{k=1}^N \delta^k = \mathcal{C}$. Recall the definition of feasible floors: a category is satisfied if it is not possible to add funding towards the highest-merit grant eligible for the category that is not yet fully funded without exceeding the floor. The approach we take to implement the *solution procedure* is simple: for the current group δ^k , iteratively fund the best grant that is not yet fully funded and is eligible for a yet-unsatisfied category in δ^k . Then, look at the next group δ^{k+1} , and so on, until all floors are satisfied; at the end, fund the best grants regardless of category. One feature of this procedure is important for understanding how it behaves: grants may be eligible for multiple categories, and when a grant is funded, it counts towards the floors of *all* categories it is eligible for. This means a late group δ^τ may have most of its floors met before its turn if earlier steps funded grants eligible for multiple categories. We define the *floors-first choice rule* C_Δ^f as the mapping from floor problems to the outcome ξ of the floors-first algorithm.

Let $\xi^0 = 0 \in \mathbb{R}^n$ be the initial vector of funding;

Step $t \geq 1$:

if ξ^{t-1} fails Axiom C for some category, let $\delta_t \in \Delta$ be the smallest-index $\delta \in \Delta$ containing such a category, and define $\mathcal{C}^t \subseteq \delta_t$ as the set of all categories $c \in \delta_t$ such that ξ^{t-1} does not satisfy Axiom C for c ; otherwise, let $\mathcal{C}^t = \emptyset$;

if the ξ^{t-1} -budget set is nonempty **then**

if $\mathcal{C}^t \neq \emptyset$ **then**

 let i_t be the highest-merit grant in $\bigcup_{c \in \mathcal{C}^t} A_c(\xi^{t-1})$, i.e. the highest-merit grant eligible for some $c \in \mathcal{C}^t$ that is not fully funded;
 set $M_t = \max_{c \in \mathcal{C}^t \cap \rho(i_t)} \{d_c(\xi^{t-1})\}$;

else

 let i_t be the highest-merit grant in $\mathcal{B}_{\xi^{t-1}}$; let $M_t = \mathcal{B} - \sum_{i \in I} \xi_i^{t-1}$;

end

 set $\xi_{i_t}^t = \xi_{i_t}^{t-1} + \min \{\bar{b}_{i_t} - \xi_{i_t}^{t-1}, M_t\}$,

$\xi^t = (\xi_1^{t-1}, \dots, \xi_{i_t-1}^{t-1}, \xi_{i_t}^t, \xi_{i_t+1}^{t-1}, \dots, \xi_n^{t-1})$;

 go to the next step;

else

 the algorithm terminates, and $\xi \equiv \xi^{t-1}$;

end

floors-first algorithm

To summarize, the floors-first choice rule uses the set $\mathcal{C}^t \subseteq \mathcal{C}$ to keep track of which categories are yet unsatisfied among the lowest-index unfulfilled group δ_t . At each step, it chooses the highest-merit grant among one of these categories that is not yet fully funded, and once $\mathcal{C}^t$ is empty, it chooses the highest-merit grants in the budget set, irrespective of categories.

Theorem 2. *Fix an ordered partition Δ . The floors-first choice rule C_Δ^f is cost-monotonic, independent of non-contending grants, and satisfies Axioms A–C (non-wastefulness, respecting priorities, and satisfying feasible floors).*

By Theorem 2, the floors-first choice rule meets every requirement we placed on floor problems: it makes full use of the budget, respects the merit order, and meets every floor in the sense of feasible floors. It is also robust in two respects. A funded grant stays funded if it asks for less, and submitting an application that was never in contention leaves the allocation unchanged. These properties hold for every ordered partition Δ , so they are invariant to the ordering and grouping of the categories, even though the allocation itself may depend on Δ .

The rule's shortcoming is that it is not monotonic in merit or categories. To see why, suppose a grant shrinks its categories or slightly lowers its merit. It may then

no longer complete a category that it previously completed. (A grant *completes* a category if it causes the category's floor to count as met while itself being unfunded, because it is the next-best grant there and is sufficiently expensive.) Thus, this deviation may lead to more funding being assigned during the phase of the choice rule when $\mathcal{C}^t \neq \emptyset$. With more funding already being assigned, the grant may beat out a more expensive but higher-merit grant in the open phase due to there now being too little funding left over for the rival grant. Observe that this story only works when the grant completes a category. However, if we only consider non-marginal grants, that cannot be the case, and it turns out that they cannot benefit from a monotonic disimprovement.

Proposition 5. *Fix an ordered partition Δ , a floor problem $\mathcal{G}$, and a grant $j \in I$ that is non-marginal at $\xi := C_\Delta^f(\mathcal{G})$. Let $\theta_j \succsim_t \theta'_j$ and let $\mathcal{G}'$ be $\mathcal{G}$ with j 's report replaced by θ'_j . Then $C_\Delta^f(\mathcal{G}') = \xi$. In particular C_Δ^f is monotonic in types for non-marginal grants.*

Corollary 1. *Let $\theta_j \succsim_t \theta'_j$. If C_Δ^f funds j when it reports θ'_j but not when it reports θ_j , then j is marginal at $\xi := C_\Delta^f(\theta_j, \theta_{-j})$; specifically, j is the next-best grant of one of its categories, the highest-merit member of some I_c with $c \in \rho(j)$ that is not fully funded.*

Proof. Were j non-marginal at ξ , Proposition 5 would give $C_\Delta^f(\theta'_j, \theta_{-j}) = \xi$, so j would be unfunded under θ'_j as well, contrary to hypothesis. Hence j is marginal at ξ . The first condition of non-marginality holds automatically: j is unfunded at ξ , and the algorithm halts only once the ξ -budget set is empty, so $\underline{b}_j > \mathcal{B} - \sum_{i \in I} \xi_i$. Therefore j fails the second condition, meaning that for some $c \in \rho(j)$ every higher-merit member of I_c is fully funded, which is the stated conclusion. $\square$

While the floors-first choice rule is not monotonic in types, we can say exactly where non-monotonicity can occur. By Corollary 1, a grant can lose funding through a monotonic improvement only if it is marginal at the resulting allocation, which here means it is the next-best grant of one of its categories; equivalently, among grants unfunded at ξ , only those next-best in some category at ξ can benefit from a monotonic disimprovement. Non-marginal grants are immune in a stronger sense: by Proposition 5, no monotonic disimprovement in their report changes the allocation at all. Moreover, the exposed population is small, and its size does not grow with the number of applicants.

Corollary 2. *At any allocation produced by C_Δ^f , at most $|\mathcal{C}|$ unfunded grants can benefit from a monotonic disimprovement, regardless of $|I|$.*

Proof. By Corollary 1, any such grant j is the highest-merit not-fully-funded grant of some $c \in \rho(j)$. That grant is unique for each category, so the assignment of an exposed grant to a category witnessing it is injective into $\mathcal{C}$. $\square$

Hence, for a large population of unfunded grants, the proportion which can benefit from a monotonic disimprovement vanishes: the number that can is capped by the number of categories, a quantity fixed by the policymaker rather than by the applicant pool.

To give a clear picture of how the choice rule works, we provide an example. To make matters simpler, we allow full funding only, so instead of using a funding vector ξ^t to indicate funding at time t , we will use a set A^t consisting of grants receiving funding at time t .

Example 3. Suppose the floor problem $\mathcal{G}$ consists of five grants $i_1 \succ i_2 \succ i_3 \succ i_4 \succ i_5$ given by:

$$\begin{aligned} i_1 : \quad b_1 &= 1, & \rho(i_1) &= \{c_A\} \\ i_2 : \quad b_2 &= 1, & \rho(i_2) &= \{c_B\} \\ i_3 : \quad b_3 &= 2, & \rho(i_3) &= \{c_A, c_B\} \\ i_4 : \quad b_4 &= 1, & \rho(i_4) &= \{c_A\} \\ i_5 : \quad b_5 &= 1, & \rho(i_5) &= \{c_B\}, \end{aligned}$$

with $\mathcal{C} = \{c_A, c_B\}$, $\mathcal{B} = 4$, $q_{c_A} = 2$, and $q_{c_B} = 2$.

Step 1 of C_Δ^f : $A^0 = \emptyset$, and $\mathcal{C}^1 = \{c_A, c_B\}$. Among grants i eligible for at least one category $c_k \in \{c_A, c_B\}$, i_1 has the highest merit. Therefore, $A^1 = \{i_1\}$.

Step 2 of C_Δ^f : Now $\mathcal{C}^2 = \{c_B\}$, since adding the next highest-merit grant in category c_A , namely i_3 , would cause funds allocated to c_A to exceed the floor. Among grants eligible for category c_B , i_2 has the highest merit, so $A^2 = \{i_1, i_2\}$.

Step 3 of C_Δ^f : Now $\mathcal{C}^3 = \emptyset$ because for either category, i_3 is the next highest-merit grant, and its addition would cause either category to have funds in excess of the floor. Hence, in this step, C_Δ^f chooses the highest-merit grant in the budget set, which happens to be i_3 . Therefore, $A^3 = \{i_1, i_2, i_3\}$.

Step 4 of C_Δ^f : Now the budget set is empty, so the algorithm terminates with $C_\Delta^f(\mathcal{G}) = \{i_1, i_2, i_3\}$.

In this section, we constructed a choice rule that follows a simple pattern of meeting floors first, then assigning open funding. This leads to a non-wasteful allocation that respects priorities and satisfies feasible floors. Yet, it is not the unique solution, so we will spend time considering a refinement.

6.3 Computation

Each step of the floors-first algorithm makes permanent progress of one of three kinds: it meets a category's floor, it fully funds a grant, or it exhausts the budget. Since there are $|\mathcal{C}|$ categories and $|I|$ grants, this bounds the number of steps.

Proposition 6. *The floors-first algorithm operates in polynomial time with a total time complexity of $\mathcal{O}((|I| + |\mathcal{C}|) \cdot |I| \cdot |\mathcal{C}|)$ and terminates in at most $|\mathcal{C}| + |I| + 1$ steps.*

7 Merit-Maximal Choice Rule

The idea of using merit to totally order allocations and selecting the maximum is due to [Sönmez and Yenmez \(2019a\)](#), where it appears implicitly. We adapt it to our setting, where allocations are continuous and the admissible set is defined by Axioms A–C, and give the construction explicitly. Fixing a floor problem, any finite set of allocations satisfying a given set of restrictions is totally ordered, as we show below, and therefore has a maximal element. For problems with earmarks, we have shown there is a unique solution satisfying the desiderata, so clearly the total order of possible allocations for each problem contains just one element. Therefore, this section focuses on floor problems, and hence also focuses on choice rules. We mentioned that this approach requires a finite set of allocations; to meet this requirement, we restrict attention to integer problems and to choice rules giving integer solutions. First, some definitions are required.

If ξ and ξ' are allocations for a problem $\mathcal{G}$, we say that ξ *dominates* ξ' if $\xi \neq \xi'$ and the highest-merit grant i^* having $\xi_{i^*} \neq \xi'_{i^*}$ satisfies $\xi_{i^*} > \xi'_{i^*}$, and denote this $\xi \triangleright \xi'$.

Proposition 7. *Fix a grant-funding problem $\mathcal{G}$. Any nonempty finite set Ξ of allocations for $\mathcal{G}$ is totally ordered by $\triangleright$ and contains a maximum ξ^* .*

Proof. If $\xi, \xi' \in \Xi$ and $\xi \neq \xi'$, then, since I is finite and $\succ$ is strict, there is a unique highest-merit grant i^* for which $\xi_{i^*} \neq \xi'_{i^*}$. Exactly one of $\xi_{i^*} > \xi'_{i^*}$ and $\xi_{i^*} < \xi'_{i^*}$ holds. In the first case $\xi \triangleright \xi'$, and in the second $\xi' \triangleright \xi$. Hence any two distinct elements of Ξ are comparable, and exactly one dominates the other; and $\triangleright$ is irreflexive by definition.

It remains to show that $\triangleright$ is transitive. Suppose $\xi \triangleright \xi'$ and $\xi' \triangleright \xi''$, with i_1 the highest-merit grant at which ξ and ξ' differ and i_2 the highest-merit grant at which ξ' and ξ'' differ. Let j be whichever of i_1, i_2 has higher merit ($j = i_1$ if $i_1 = i_2$). Every grant of higher merit than j has higher merit than both i_1 and i_2 , so ξ , ξ' and ξ'' all assign it the same amount. At j itself, $\xi_j \geq \xi'_j \geq \xi''_j$, and at least one of these inequalities is strict: if $j = i_1$, then $\xi_j > \xi'_j$ and $\xi'_j \geq \xi''_j$ (with equality unless $j = i_2$);

if $j = i_2 \neq i_1$, then $\xi_j = \xi'_j > \xi''_j$. Thus $\xi_j > \xi''_j$, so $\xi \neq \xi''$, j is the highest-merit grant at which ξ and ξ'' differ, and $\xi \succ \xi''$.

Therefore $\succ$ is a strict total order on Ξ . Since Ξ is finite and nonempty, it has a maximum (by induction on $|\Xi|$), and the maximum is unique by asymmetry. In other words, there exists a unique $\xi^* \in \Xi$ such that $\xi^* \succ \xi$ for all $\xi \in \Xi \setminus \{\xi^*\}$. $\square$

We call a grant-funding problem $\mathcal{G}$ an *integer problem* if each item in the tuple is an integer, and we call a choice rule *integer-valued* if it assigns integer values of funding to each grant. Suppose $\tilde{\mathcal{M}}$ is a non-empty collection of integer-valued choice rules defined on a collection of integer problems Γ . Then we say that C is *merit-maximal on $\tilde{\mathcal{M}}$* if C belongs to $\tilde{\mathcal{M}}$ and for every choice rule $C' \in \tilde{\mathcal{M}}$ having $C' \neq C$, (1) for every $\mathcal{G} \in \Gamma$, $C'(\mathcal{G})$ does not dominate $C(\mathcal{G})$, and (2) there exists an integer problem $\mathcal{G} \in \Gamma$ such that $C(\mathcal{G})$ dominates $C'(\mathcal{G})$. Equivalently, a merit-maximal choice rule is *constrained lexicographically efficient*: among the allocations meeting the imposed criteria, it selects the one that is largest in the lexicographic order $\succ$ induced by merit.

To motivate this approach, if one considers the primary purpose of the floors-first choice rule as finding a choice that satisfies a list of criteria (Axioms A–C), then if we codify the set of choice rules satisfying those criteria as $\mathcal{M}$, the merit-maximal choice rule on $\mathcal{M}$ is in some sense the best choice rule among $\mathcal{M}$, whereas C_Δ^f is merely *some* choice rule belonging to $\mathcal{M}$. We will show that this reasoning is not foolproof, as there is a drawback to the merit-maximal choice rule on $\mathcal{M}$; namely, such a rule cannot be monotonic in cost, merit, or categories, nor independent of non-contending grants.

7.1 Application to Axioms A–C

Let Γ be the set of all integer floor problems, and for each $\mathcal{G} \in \Gamma$, let $\Xi(\mathcal{G})$ be the set of integer allocations ξ satisfying Axioms A–C (non-wastefulness, respecting priorities, and satisfying feasible floors). Then, $\mathcal{M} = \{C : \mathcal{G} \mapsto \xi \in \Xi(\mathcal{G}) \text{ defined on } \Gamma\}$ is the collection of integer-valued choice rules defined on integer problems without earmarks that give allocations satisfying Axioms A–C.

Proposition 8. *For any Δ , C_Δ^f belongs to $\mathcal{M}$.*

Proof. Assume there is an integer floor problem $\mathcal{G} \in \Gamma$. Fix a step t of C_Δ^f and assume that ξ^t is an integer vector; hence, the amount of budget remaining is an integer and each floor shortfall $q_c - \sum_{i \in I_c} \xi_i^t$ is an integer value. In step $t + 1$, compared to ξ^t the tentative allocation ξ^{t+1} either meets a floor q_c exactly, fully funds a grant, or exhausts the budget. In any of these cases, an integer value of funding is assigned. Hence, ξ^{t+1} is an integer vector. Clearly $\xi^0 = \vec{0}$ is an integer vector, so it follows by induction that $C_\Delta^f(\mathcal{G})$ is an integer vector. The result follows. $\square$

Proposition 9. *The choice rule $C_{\mathcal{M}}^{\text{mm}}$ that assigns to each $\mathcal{G} \in \Gamma$ the $\triangleright$ -maximum of $\Xi(\mathcal{G})$ is well defined, and it is the unique merit-maximal choice rule on $\mathcal{M}$.*

Proof. For each $\mathcal{G} \in \Gamma$, one may define $C_{\mathcal{M}}^{\text{mm}}(\mathcal{G}) = \xi^*$ where ξ^* is the maximum of $\Xi(\mathcal{G})$; this is possible by Proposition 7, since $\Xi(\mathcal{G})$ is finite and, by Proposition 8, nonempty. It is clear that $C_{\mathcal{M}}^{\text{mm}} \in \mathcal{M}$ because $\xi^* \in \Xi(\mathcal{G})$ for each $\mathcal{G} \in \Gamma$. It is merit-maximal on $\mathcal{M}$: for any $C' \in \mathcal{M}$ with $C' \neq C_{\mathcal{M}}^{\text{mm}}$, no $C'(\mathcal{G})$ dominates $C_{\mathcal{M}}^{\text{mm}}(\mathcal{G})$, since $C'(\mathcal{G}) \in \Xi(\mathcal{G})$ and $C_{\mathcal{M}}^{\text{mm}}(\mathcal{G})$ is the maximum of $\Xi(\mathcal{G})$; and $C'(\mathcal{G}) \neq C_{\mathcal{M}}^{\text{mm}}(\mathcal{G})$ for some $\mathcal{G} \in \Gamma$, where therefore $C_{\mathcal{M}}^{\text{mm}}(\mathcal{G}) \triangleright C'(\mathcal{G})$. It is the only such rule: if $C \in \mathcal{M}$ with $C \neq C_{\mathcal{M}}^{\text{mm}}$, then $C(\mathcal{G}) \neq C_{\mathcal{M}}^{\text{mm}}(\mathcal{G})$ for some $\mathcal{G} \in \Gamma$, and there $C_{\mathcal{M}}^{\text{mm}}(\mathcal{G}) \triangleright C(\mathcal{G})$, so C violates condition (1) of the definition with $C' = C_{\mathcal{M}}^{\text{mm}}$. $\square$

To gain a better understanding of $C_{\mathcal{M}}^{\text{mm}}$, we consider an example in which $C_{\mathcal{M}}^{\text{mm}}$ and C_{Δ}^f agree. As usual, we use a set rather than a vector to denote the chosen grants, since $\bar{b}_i = \bar{b}_i$ for all i .

Example 4. *Consider the grant-funding problem where grants $i_1 \succ i_2 \succ i_3 \succ i_4 \succ i_5$ are as follows:*

$$\begin{aligned} i_1 : \quad & b_1 = 13, \quad \rho(i_1) = \emptyset \\ i_2 : \quad & b_2 = 5, \quad \rho(i_2) = \{c\} \\ i_3 : \quad & b_3 = 2, \quad \rho(i_3) = \{c\} \\ i_4 : \quad & b_4 = 2, \quad \rho(i_4) = \{c\} \\ i_5 : \quad & b_5 = 5, \quad \rho(i_5) = \{c\}, \end{aligned}$$

where $\mathcal{C} = \{c\}$, $\mathcal{B} = 20$, and $q_c = 10$.

We first find $C_{\mathcal{M}}^{\text{mm}}$. Consider the attainable subsets of I (that is, the attainable allocations) that contain i_1 . If any of them satisfies Axioms A–C, then $C_{\mathcal{M}}^{\text{mm}}$ funds i_1 ; otherwise, it does not. Since $\mathcal{B} - b_1 = 7$, a subset containing i_1 is attainable exactly when the other grants in it cost at most 7 in total, which gives ten such subsets. Five of them— $\{i_1\}$, $\{i_1, i_2\}$, $\{i_1, i_3\}$, $\{i_1, i_4\}$, and $\{i_1, i_5\}$ —fail Axiom A: each leaves at least 2 of the budget unspent, so adding i_3 or i_4 , whichever it lacks, yields a larger attainable allocation. Of the other five, $\{i_1, i_2, i_3\}$, $\{i_1, i_2, i_4\}$, and $\{i_1, i_3, i_4\}$ fail Axiom C, while $\{i_1, i_3, i_5\}$ and $\{i_1, i_4, i_5\}$ fail Axiom B. Hence no attainable subset containing i_1 satisfies Axioms A–C.

Had Axiom B been omitted from the definition of $\mathcal{M}$, $C_{\mathcal{M}}^{\text{mm}}$ would have selected $\{i_1, i_3, i_5\}$. This would be undesirable, because i_2 is a strictly better grant than i_5 : it has the same cost and the same categories, but *higher* merit. Axiom B rules out this outcome, and as a result $C_{\mathcal{M}}^{\text{mm}}$ does not fund i_1 . Exactly one subset that excludes i_1 satisfies Axioms A–C, namely $\{i_2, i_3, i_4, i_5\}$, so $C_{\mathcal{M}}^{\text{mm}}(\mathcal{G}) = \{i_2, i_3, i_4, i_5\}$.

For C_{Δ}^f , we must have $\Delta = \{\{c\}\}$, and i_2 is chosen in the first step; the remaining steps give $C_{\Delta}^f(\mathcal{G}) = C_{\mathcal{M}}^{\text{mm}}(\mathcal{G}) = \{i_2, i_3, i_4, i_5\}$. The two rules agree in this example, but they do not agree in general.

7.2 Limits of Merit-Maximality

Recall, we defined $\mathcal{M} = \{C : \mathcal{G} \mapsto \xi \in \Xi(\mathcal{G}) \text{ defined on } \Gamma\}$ as the collection of integer-valued choice rules defined on integer floor problems that give allocations satisfying Axioms A–C (non-wasteful, respects priorities, satisfies feasible floors).

Theorem 3. *Any choice rule $C \in \mathcal{M}$ which is independent of non-contending grants, or monotonic in cost, merit, or categories, or which is monotonic in types for non-marginal grants is not merit-maximal on $\mathcal{M}$.*

Any such choice rule $C \in \mathcal{M}$ must sometimes lead to outcomes which are dominated by another outcome satisfying Axioms A–C. In practice, this means that a policymaker who wants to reward better applications, or to keep the allocation unaffected by the arrival of applications that are out of contention, must give up efficiency with respect to the total order.

First, we show that $C_{\mathcal{M}}^{\text{mm}}$ fails monotonicity in cost at a non-marginal grant, so it is neither monotonic in cost nor monotonic in types for non-marginal grants.

Example 5. *Let $\mathcal{C} = \{c\}$, $\mathcal{B} = 16$ and $q_c = 12$, and let there be three grants $i_1 \succ i_2 \succ i_3$ given by*

$$\begin{aligned} i_1 : [\underline{b}_1, \bar{b}_1] &= [1, 10], & \rho(i_1) &= \emptyset \\ i_2 : [\underline{b}_2, \bar{b}_2] &= [7, 13], & \rho(i_2) &= \{c\} \\ i_3 : [\underline{b}_3, \bar{b}_3] &= [2, 4], & \rho(i_3) &= \{c\}. \end{aligned}$$

So $I_c = \{i_2, i_3\}$, and every component of $\mathcal{G}$ is an integer.

The merit-maximal allocation is $C_{\mathcal{M}}^{\text{mm}}(\mathcal{G}) = (4, 12, 0)$. However, when i_3 deviates by increasing its cost to $[2, 6]$, the merit-maximal allocation is $C_{\mathcal{M}}^{\text{mm}}(\mathcal{G}') = (10, 0, 6)$. Therefore, i_3 benefits using a monotonic disimprovement. Intuitively, if enough funding may be allocated to i_3 , then it is possible that i_2 , the next-best grant in the category, is too expensive to be added without exceeding the floor. Therefore, the feasible floor is satisfied for c , and there is still more funding left over to give to i_1 , so after the deviation, the new allocation dominates the previous one. (A detailed proof is given in Appendix A.3.)

Because every choice rule we consider depends on merit only through the order $\succ$, and never through the cardinal values m_i , this deviation admits two readings, either merit falls or stays the same while the request is enlarged. Either reading suffices for Theorem 3.

Next, we illustrate that neither C_{Δ}^f nor $C_{\mathcal{M}}^{\text{mm}}$ is monotonic in merit or categories:

Example 6. Suppose there are three grants $i_1 \succ i_2 \succ i_3$, given by:

$$\begin{aligned} i_1 : \quad & b_1 = 4, \quad \rho(i_1) = \emptyset \\ i_2 : \quad & b_2 = 2, \quad \rho(i_2) = \{c\} \\ i_3 : \quad & b_3 = 1, \quad \rho(i_3) = \{c\}, \end{aligned}$$

where $\mathcal{C} = \{c\}$, $\mathcal{B} = 4$, and $q_c = 1$. Further, it must be that $\Delta = \{\{c\}\}$.

Notice that $C_\Delta^f(\mathcal{G}) = C_{\mathcal{M}}^{\text{mm}}(\mathcal{G}) = \{i_1\}$. However, if we make the adjustment $\rho(i_2) = \emptyset$ and call the resulting problem $\mathcal{G}'$, then $C_\Delta^f(\mathcal{G}') = C_{\mathcal{M}}^{\text{mm}}(\mathcal{G}') = \{i_2, i_3\}$, so i_2 can benefit from failing to report category c . If instead we lower the merit of i_2 so that $i_1 \succ i_3 \succ i_2$ and call the resulting problem $\mathcal{G}''$, then $C_\Delta^f(\mathcal{G}'') = C_{\mathcal{M}}^{\text{mm}}(\mathcal{G}'') = \{i_2, i_3\}$, so lowering the merit of i_2 turns its rejection into an award.

Finally, we show $C_{\mathcal{M}}^{\text{mm}}$ is not independent of non-contending grants.

Example 7. Suppose there are three grants $i_1 \succ i_2 \succ j$, given by:

$$\begin{aligned} i_1 : \quad & [b_1, \bar{b}_1] = [1, 2], \quad \rho(i_1) = \emptyset \\ i_2 : \quad & [b_2, \bar{b}_2] = [2, 3], \quad \rho(i_2) = \{c\} \\ j : \quad & [b_j, \bar{b}_j] = [1, 1], \quad \rho(j) = \{c\}, \end{aligned}$$

where $\mathcal{C} = \{c\}$, $\mathcal{B} = 3$, and $q_c = 2$.

In the problem with only $I = \{i_1, i_2\}$, the allocation is $C_{\mathcal{M}}^{\text{mm}}(\mathcal{G}) = (1, 2)$, whereas in the augmented problem, $\mathcal{G}^{+j}$, the allocation is $C_{\mathcal{M}}^{\text{mm}}(\mathcal{G}^{+j}) = (2, 0, 1)$. Hence, the rule fails independence of non-contending grants. Grant j is non-marginal at $C_{\mathcal{M}}^{\text{mm}}(\mathcal{G}) = (1, 2)$: its cost exceeds the leftover funds, which are zero, and in its only category the higher-merit grant i_2 is funded below its ceiling. It is therefore non-contending there, since i_1 and i_2 are the only funded grants and both rank above j .

Corollary 3. For any Δ , C_Δ^f is not merit-maximal on $\mathcal{M}$.

Proof. By Proposition 8, $C_\Delta^f \in \mathcal{M}$, and by Theorem 2, C_Δ^f is monotonic in cost. Theorem 3 then implies that C_Δ^f is not merit-maximal on $\mathcal{M}$. $\square$

The merit order determines a unique merit-maximal choice rule on $\mathcal{M}$, namely $C_{\mathcal{M}}^{\text{mm}}$ (Proposition 9); however, $C_{\mathcal{M}}^{\text{mm}}$ is neither monotonic in cost, merit, or categories nor independent of non-contending grants. By Theorem 3, no choice rule in $\mathcal{M}$ that is monotonic in cost or independent of non-contending grants is merit-maximal on $\mathcal{M}$; merit-maximality must therefore be relinquished to obtain either property. The floors-first choice rule is not merit-maximal (Corollary 3), but it satisfies both properties (Theorem 2). These properties, together with its transparency and ease of implementation, may lead a policymaker to prefer C_Δ^f to $C_{\mathcal{M}}^{\text{mm}}$.

7.3 Computation

The merit-maximal choice rule selects the maximum of a set of allocations whose size grows exponentially in $|I|$, and we know of no implementation of it that avoids enumerating that set. The floors-first choice rule, by contrast, is implementable in polynomial time. This is a further respect, alongside monotonicity in cost and independence of non-contending grants, in which C_{Δ}^f may be preferred to $C_{\mathcal{M}}^{\text{mm}}$ in practice.

8 Conclusion

Priority design extends from indivisible assignment problems to settings where divisible funds are allocated to recipients that have minimum funding requirements and may be eligible for many funding categories. When funding is earmarked, we characterize the solution by requiring that eligible categories contribute equally to a grant wherever they can. Floors, on the other hand, cannot always be met in this setting, so we work with a relaxed notion, feasible floors. Conceptually, a floor problem does not designate which funds count towards which category, and the relaxed notion of floors admits a wider set of acceptable allocations; together these introduce enough degrees of freedom that a unique characterization is not attained.

We therefore turn to a refinement. The merit-maximal choice rule, which is constrained lexicographically efficient for integer floor problems, is unique, but relative to the floors-first choice rule it has inferior incentive properties and is sensitive to the arrival of grants that were never in contention for funding. Since efficiency comes at this cost, we recommend the floors-first choice rule: it is cost-monotonic, it is monotonic in types except at marginal grants, whose number is bounded by the number of categories, and it is independent of non-contending grants. Its appeal is also conceptual, since it settles every funding decision at one of the two margins identified in Section 4.5: the budget margin, funds left idle, and the merit frontier, the best grant not yet fully funded within each category and overall. Like the serial-SR choice rule, it can be implemented in polynomial time.

Which scheme is appropriate depends on the institution. Where funding carries strict designations—a donor who restricts how a gift may be spent, or a line item appropriated for a stated purpose—earmarks are the right instrument, and the cost of a grant must be apportioned among the categories that support it. Where instead a policymaker wants a balanced portfolio, and the setting permits a funded project to count in full towards every category it is eligible for, floors are the appropriate constraint.

Our model takes the merit ranking as given rather than modeling the two-tiered

review that produces it, and it treats reported cost bounds as observable and the allocation as one-shot. Relaxing these assumptions, and incorporating other institutional features, suggests several directions for future research. One is “soft reserves,” in which funds are set aside for a category but may be used for any grant once every grant in that category has been funded. Another is dynamic allocation, in which a rule must weigh the opportunity cost of committing funds now against waiting for later applications. Finally, two questions about floor problems remain open: whether Axioms A, B, and C^* can always be satisfied simultaneously (Section A.4), and whether the choice rules for floor problems admit an axiomatic characterization. Minimum funding requirements considerably complicate the analysis, but they supply the realism that budgeting in practice demands. More broadly, our results show that the tools of priority design extend to the allocation of divisible funds under minimum funding requirements, and we hope they provide a foundation for further work on grant funding.

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A Proofs

A.1 Earmarks

Recall that i_t denotes the grant considered at step t of the serial-SR algorithm, and that x^t is the intermediate matrix it has produced after that step.

Lemma 1. *If $\sum_{c \in \rho(i_t)} (q_c - x_c^{t-1}) < \underline{b}_{i_t}$, then $x_{i_t} = 0$. Otherwise $x_{i_t} \geq \underline{b}_{i_t}$.*

Proof of Lemma 1. If $\sum_{c \in \rho(i_t)} (q_c - x_c^{t-1}) < \underline{b}_{i_t}$, $x_{i_t}^t = 0$ by construction, and $x_{i_t} = x_{i_t}^t$. Given $\sum_{c \in \rho(i_t)} (q_c - x_c^{t-1}) \geq \underline{b}_{i_t}$, $x_{i_t}^t$ is chosen to solve the equation $x_{i_t}^t = \min\{\bar{b}_{i_t}, \sum_{c \in \rho(i_t)} (q_c - x_c^{t-1})\}$. But, the assumption implies $\min\{\bar{b}_{i_t}, \sum_{c \in \rho(i_t)} (q_c - x_c^{t-1})\} \geq \min\{\bar{b}_{i_t}, \underline{b}_{i_t}\} = \underline{b}_{i_t}$, so $x_{i_t} = x_{i_t}^t \geq \underline{b}_{i_t}$. $\square$

We use Lemma 1 in the proofs of Theorem 1, Proposition 2 and Proposition 3.

Proof of Theorem 1.

Claim 1.1. *At each step t , the serial-SR algorithm is well-defined. Moreover, the algorithm terminates in finite time, and its outcome is a matrix x .*

Proof. If $\sum_{c \in \rho(i_t)} (q_c - x_c^{t-1}) < \underline{b}_{i_t}$, then $x_{i_t}^t = 0$ is well defined. Thus, assume $\sum_{c \in \rho(i_t)} (q_c - x_c^{t-1}) \geq \underline{b}_{i_t}$. We now show that (1) and (2) admit a unique solution $\{x_{i_t,c}^t\}_{c \in \rho(i_t)}$. Let $g_c(s) = \min\{s, q_c - x_c^{t-1}\}$, and let $g(s) = \sum_{c \in \rho(i_t)} g_c(s)$. Notice that $g(0) = 0$, and at $\bar{s} = \max_{c \in \rho(i_t)} (q_c - x_c^{t-1})$, g attains its maximum $g(\bar{s}) = \sum_{c \in \rho(i_t)} (q_c - x_c^{t-1})$. Further, g is continuous and increasing on $[0, \bar{s})$, and constant on $[\bar{s}, \infty)$, so it attains the intermediate value $g(s^*) = \min\{\bar{b}_{i_t}, \sum_{c \in \rho(i_t)} (q_c - x_c^{t-1})\}$ uniquely at some $s^* \in [0, \bar{s}]$ and thus satisfies (1). In addition, for c that satisfy $s^* \leq q_c - x_c^{t-1}$, $g_c(s^*) = s^* = \max_{c' \in \rho(i_t)} \{x_{i_t,c'}^t\}$; and, for c such that $s^* > q_c - x_c^{t-1}$, $g_c(s^*) = q_c - x_c^{t-1}$. Thus, $g_c(s^*)$ satisfies (2).

For the second claim, consider that each step corresponds to a distinct grant, and there are finitely many grants. It follows there is a last step n that considers the grant i_n where $n = |I|$. $\square$

Claim 1.2. *If $\underline{b}_{i_t} \leq x_{i_t} < \bar{b}_{i_t}$, then $q_c - x_c^t = 0$ for all $c \in \rho(i_t)$.*

Proof. The assumption implies $x_{i_t} = \sum_{c \in \rho(i_t)} (q_c - x_c^{t-1})$, as x_{i_t} satisfies (1). By definition $x_{i_t} = \sum_{c \in \rho(i_t)} x_{i_t,c}$, and by (2) $x_{i_t,c} \leq q_c - x_c^{t-1}$, for all c . Therefore, $x_{i_t,c} = q_c - x_c^{t-1}$, for all $c \in \rho(i_t)$. Since $x_c^t = x_c^{t-1} + x_{i_t,c}$, the result follows. $\square$

Claim 1.3. *x is category-neutral.*

Proof. Consider any grant i_t . If $x_{i_t} = 0$ then

$$\sum_{c \in \rho(i_t)} (q_c - x_c^{t-1}) < \underline{b}_{i_t} \implies \sum_{c \in \rho(i_t)} \left(q_c - \sum_{i' \in \hat{I}_{i_t,c}} x_{i',c} \right) < \underline{b}_{i_t}.$$

Now suppose $x_{i_t} > 0$. Any funding assigned before step t is assigned to higher-merit grants within the same category from which the funds are assigned. Therefore, if $x_{i_t,c} < \max_{c' \in \rho(i_t)} \{x_{i_t,c'}^t\}$, (2) implies $x_{i_t,c} = q_c - x_c^{t-1}$, which is equivalent to $x_{i_t,c} + \sum_{i' \in \hat{I}_{i_t,c}} x_{i',c} = q_c$. $\square$

Claim 1.4. *x is eligibility-compliant.*

Proof. By construction, $x_{i,c} > 0$ only if $c \in \rho(i)$. $\square$

Claim 1.5. *x is non-wasteful.*

Proof. By Claim 1.4, x is eligibility-compliant, and Axiom A compares it only against eligibility-compliant matrices; the funds available to a grant i are accordingly the unassigned reserves in $\rho(i)$. Suppose for some t , there is i_t having $x_{i_t} = 0$. Then, by construction, $\sum_{c \in \rho(i_t)} (q_c - x_c^{t-1}) < \underline{b}_{i_t}$. Instead suppose $\bar{b}_{i_t} > x_{i_t} > 0$. It must be that $\sum_{c \in \rho(i_t)} (q_c - x_c^{t-1}) \geq \bar{b}_{i_t}$, as the opposite implies $x_{i_t} = 0$. Now, (1) implies $x_{i_t} = \min\{\bar{b}_{i_t}, \sum_{c \in \rho(i_t)} (q_c - x_c^{t-1})\} = \sum_{c \in \rho(i_t)} (q_c - x_c^{t-1})$, as $x_{i_t} < \bar{b}_{i_t}$. $\square$

Claim 1.6. x respects priorities.

Proof. Fix $i_t \in I$ and assume $x_{i_t} = 0$. The algorithm leaves i_t unfunded only when $\sum_{c \in \rho(i_t)} (q_c - x_c^{t-1}) < \underline{b}_{i_t}$, since otherwise Lemma 1 would give $x_{i_t} \geq \underline{b}_{i_t}$. Now take any $J \subseteq I$. If $i_t \pi_c j$ for some $j \in J$ and some category c , then i_t and j are both eligible for c and $i_t \succ j$, so $c \in \rho(i_t)$ and j is considered after step t ; every unit counted by $\sum_{j \in J} \sum_{c: i_t \pi_c j} x_{j,c}$ is therefore drawn from reserves in $\rho(i_t)$ that stood unassigned at the end of step $t-1$. The reserves $\sum_{c \in \rho(i_t)} (q_c - x_c)$ left unassigned at the end are drawn from that same pool and are disjoint from those units. Hence

$$\sum_{j \in J} \sum_{c: i_t \pi_c j} x_{j,c} + \sum_{c \in \rho(i_t)} (q_c - x_c) \leq \sum_{c \in \rho(i_t)} (q_c - x_c^{t-1}) < \underline{b}_{i_t},$$

so there is no J of the kind the first clause forbids.

Instead assume $\underline{b}_{i_t} \leq x_{i_t} < \bar{b}_{i_t}$. Assume there is $i_{t'}$ having $i_t \succ i_{t'}$ and $\sum_{c \in \rho(i_t)} x_{i_{t'},c} > 0$, so that $t' > t$. It follows there exists $c \in \rho(i_t)$ with $x_{i_{t'},c} > 0$. By construction, $x_{i_{t'},c} = \min\{q_c - x_c^{t'-1}, \max_{c' \in \rho(i_{t'})} \{x_{i_{t'},c'}^{t'}\}\} \leq \min\{q_c - x_c^t, \max_{c' \in \rho(i_{t'})} \{x_{i_{t'},c'}^{t'}\}\}$, since $x_c^{\tilde{t}}$ is monotonic in $\tilde{t}$. But $q_c - x_c^t = 0$ by Claim 1.2. Thus, $x_{i_{t'},c} = \min\{0, \max_{c' \in \rho(i_{t'})} \{x_{i_{t'},c'}^{t'}\}\} = 0$, a contradiction. $\square$

Claim 1.7. Any disaggregate allocation x' that is eligibility-compliant, category-neutral, non-wasteful, and respects priorities must coincide with x .

Proof. Assume $x' \neq x$ is eligibility-compliant and non-wasteful. We will show this assumption either leads to a contradiction, or implies x fails one of the other axioms. Eligibility-compliance allows us to assume all funding towards i in x' is from categories in $\rho(i)$, an assumption we will make use of throughout below.

Since $x' \neq x$, some row differs: $x'_{i,c} \neq x_{i,c}$ for some i and c . Let i_t be the highest-merit grant for which this holds, and let $\hat{I}_{i_t} = \{i \in I : i \succ i_t\}$. By the choice of i_t , $x'_{i,c} = x_{i,c}$ for every $i \in \hat{I}_{i_t}$ and every c ; that is, the higher-merit grants agree with x category by category, not merely in their totals. Writing $x_c^{t'-1} = \sum_{i \in \hat{I}_{i_t}} x'_{i,c}$, and recalling that $x_c^{t-1} = \sum_{i \in \hat{I}_{i_t}} x_{i,c}$ by construction, it follows that

$$x_c^{t'-1} = x_c^{t-1} \quad \text{for all } c. \quad (4)$$

Both cases below rest on (4).

Case 1. Assume that $x'_{i_t} \neq x_{i_t}$.

Suppose $x'_{i_t} > x_{i_t}$. By definition, x_{i_t} satisfies (1). If $x_{i_t} = \bar{b}_{i_t}$, then $x'_{i_t} > x_{i_t} = \bar{b}_{i_t}$ implies x' is unattainable. Otherwise, $x'_{i_t} > \sum_{c \in \rho(i_t)} (q_c - x_c^{t-1})$: if $x_{i_t} > 0$, then $x_{i_t} = \sum_{c \in \rho(i_t)} (q_c - x_c^{t-1})$ by (1); if $x_{i_t} = 0$, then $\sum_{c \in \rho(i_t)} (q_c - x_c^{t-1}) < \underline{b}_{i_t} \leq x'_{i_t}$ by Lemma 1 and attainability of x' . Thus there is a category $c \in \rho(i_t)$ such that $x'_{i_t,c} > q_c - x_c^{t-1} = q_c - x_c^{t-1}$ by (4), so $x_c^{t-1} > q_c$, implying x' is unattainable, contradicting the assumption that x' is an allocation.

Suppose instead $x'_{i_t} < x_{i_t}$. Notice that $\underline{b}_{i_t} \leq x_{i_t} \leq \sum_{c \in \rho(i_t)} (q_c - x_c^{t-1})$. This gives $\underline{b}_{i_t} \leq x_{i_t} \leq \sum_{c \in \rho(i_t)} (q_c - \sum_{i \in \hat{I}_{i_t}} x'_{i,c})$, by (4). Let $\{j_1, \dots, j_m\}$ be the set of grants inferior in merit to i_t . By eligibility-compliance, and since $i_t \pi_c j_k$ holds exactly when $c \in \rho(i_t) \cap \rho(j_k)$, we have $\sum_{c \in \rho(i_t)} (q_c - \sum_{i \in \hat{I}_{i_t}} x'_{i,c}) = x'_{i_t} + \sum_{k=1}^m \sum_{c \in \{\bar{c} : i_t \pi_{\bar{c}} j_k\}} x'_{j_k,c} + \sum_{c \in \rho(i_t)} (q_c - x'_c)$, which implies

$$\underline{b}_{i_t} \leq x_{i_t} \leq x'_{i_t} + \sum_{k=1}^m \sum_{c \in \{\bar{c} : i_t \pi_{\bar{c}} j_k\}} x'_{j_k,c} + \sum_{c \in \rho(i_t)} (q_c - x'_c). \quad (5)$$

If $x'_{i_t} = 0$, the leading term on the right of (5) vanishes; consider the two sums that remain. Should the first vanish, the second alone is at least $\underline{b}_{i_t}$, so i_t could be funded out of reserves in $\rho(i_t)$ that sit unassigned, and x' fails to be non-wasteful. Otherwise the first sum is positive and (5) is exactly the configuration the first clause of Axiom B forbids, so x' fails to respect priorities. Hence, suppose $x'_{i_t} > 0$. Since $x'_{i_t} < x_{i_t}$, (5) implies $0 < x_{i_t} - x'_{i_t} \leq \sum_{k=1}^m \sum_{c \in \{\bar{c} : i_t \pi_{\bar{c}} j_k\}} x'_{j_k,c} + \sum_{c \in \rho(i_t)} (q_c - x'_c)$, so one of the two sums must be positive. Moreover $\underline{b}_{i_t} \leq x'_{i_t} < x_{i_t} \leq \bar{b}_{i_t}$, by attainability of x' . If the second sum is positive, then some category in $\rho(i_t)$ holds unassigned reserve, so x'_{i_t} can be raised by drawing on that category alone; the resulting matrix is eligibility-compliant and attainable, so x' fails to be non-wasteful, a contradiction. Hence, the first sum is positive, so there exist k and $c \in \rho(i_t)$ with $i_t \succ j_k$ and $x'_{j_k,c} > 0$. Since $\underline{b}_{i_t} \leq x'_{i_t} < \bar{b}_{i_t}$, x' fails the second clause of Axiom B.

Case 2. Instead, assume $x'_{i_t} = x_{i_t}$. By the choice of i_t , it still holds that $x'_{i_t,c} \neq x_{i_t,c}$ for some c . Notice that there must exist c' with $x'_{i_t,c'} > x_{i_t,c'}$, and that $x_{i_t} > 0$: otherwise $x'_{i_t} = x_{i_t} = 0$ would force $x'_{i_t,c} = x_{i_t,c} = 0$ for every c . Hence $x_{i_t} \geq \underline{b}_{i_t}$ by Lemma 1. If $x_{i_t} < \bar{b}_{i_t}$, then by Claim 1.2, $x_{i_t,c'} = q_{c'} - x_{c'}^{t-1}$. Therefore, $x'_{i_t,c'} > x_{i_t,c'}$, which leads to $x'_{i_t,c'} > q_{c'} - x_{c'}^{t-1} = q_{c'} - x_{c'}^{t-1}$ by (4). But, this implies $x_{c'}^{t-1} > q_{c'}$, and as such $x_{c'}^{t-1} > q_{c'}$, implying x' is unattainable, a contradiction.

Therefore, assume $x_{i_t} = \bar{b}_{i_t}$. Let $\mathcal{K}$ be the set of categories c for which $x_{i_t,c} = q_c - x_c^{t-1}$. Let $\mathcal{L}$ be the set of categories c for which $x_{i_t,c} < q_c - x_c^{t-1}$, so $x_{i_t,c} = \max_{\bar{c}} \{x_{i_t,\bar{c}}\}$. Notice that $\mathcal{K} \cup \mathcal{L} = \rho(i_t)$. If $x'_{i_t,c} \leq x_{i_t,c}$ for all $c \in \mathcal{L}$, then because $\sum_c x'_{i_t,c} = \sum_c x_{i_t,c}$, and there is a difference between some of the summands by

assumption, some categories in $\rho(i_t) \setminus \mathcal{L} = \mathcal{K}$ must be greater in the left sum than their right sum counterparts. In other words, $\exists \tilde{c} \in \mathcal{K}$ such that $x'_{i_t, \tilde{c}} > x_{i_t, \tilde{c}} = q_{\tilde{c}} - x_{\tilde{c}}^{t-1}$, implying x' is unattainable, a contradiction. Hence assume $\exists c' \in \mathcal{L}$ with $x'_{i_t, c'} > x_{i_t, c'}$. Then, for all $\tilde{c} \in \mathcal{L}$, $x'_{i_t, \tilde{c}} > x_{i_t, \tilde{c}}$ as otherwise would contradict category-neutrality. But this implies $\exists \tilde{c} \in \mathcal{K}$ such that $x'_{i_t, \tilde{c}} < x_{i_t, \tilde{c}} = q_{\tilde{c}} - x_{\tilde{c}}^{t-1}$, and since $c' \in \mathcal{L}$ gives $x_{i_t, \tilde{c}} \leq \max_c \{x_{i_t, c}\} = x_{i_t, c'}$, we have $x'_{i_t, \tilde{c}} < x_{i_t, \tilde{c}} \leq x_{i_t, c'} < x'_{i_t, c'} \leq \max_c \{x'_{i_t, c}\}$. Because also $x'_{i_t, \tilde{c}} + x_{\tilde{c}}^{t-1} < q_{\tilde{c}}$ by (4), x' is not category-neutral. $\square$

Claims 1.1–1.7 establish that C^{SR} is well defined and that it yields the unique allocation satisfying Axioms A, B, D and E. $\square$

Proof of Proposition 1. In each case o denotes the open category and c_A a reserved one, and grants are listed in merit order.⁴

Axiom A. Let $I = \{i_1\}$ and $\mathcal{C} = \{o\}$ with $q_o = 2 = \mathcal{B}$, where $\rho(i_1) = \{o\}$ and $[\underline{b}_{i_1}, \bar{b}_{i_1}] = [1, 2]$. Take $x_{i_1, o} = 1$. Eligibility-compliance is immediate, and with a single eligible category Axiom E is vacuous. The one grant has nothing ranked below it, so both clauses of Axiom B hold vacuously. But $x'_{i_1, o} = 2$ is attainable and eligibility-compliant with $x' \geq x$, so Axiom A fails.

Axiom B. Let $I = \{i_1, i_2\}$ with $i_1 \succ i_2$ and $\mathcal{C} = \{o\}$ with $q_o = 2 = \mathcal{B}$, where $\rho(i_1) = \rho(i_2) = \{o\}$ and $[\underline{b}_i, \bar{b}_i] = [2, 2]$ for both. Take $x_{i_2, o} = 2$ and $x_{i_1, o} = 0$. The reserve is exhausted, so no grant can absorb more and Axiom A holds; compliance is immediate and Axiom E is again vacuous. But i_1 is unfunded while i_2 , which it outranks in π_o , holds $2 \geq \underline{b}_{i_1}$, so the first clause of Axiom B fails.

Axiom D. Let $I = \{i_1\}$ and $\mathcal{C} = \{o, c_A\}$ with $q_o = q_{c_A} = 1$ and $\mathcal{B} = 2$, where $\rho(i_1) = \{o\}$ and $[\underline{b}_{i_1}, \bar{b}_{i_1}] = [2, 2]$. Take $x_{i_1, o} = x_{i_1, c_A} = 1$. Both reserves are exhausted and i_1 is funded to its ceiling, so Axioms A and B hold, and Axiom E quantifies over $\rho(i_1) = \{o\}$ alone and is vacuous. But $c_A \notin \rho(i_1)$ while $x_{i_1, c_A} > 0$, so Axiom D fails.

Axiom E. Let $I = \{i_1\}$ and $\mathcal{C} = \{o, c_A\}$ with $q_o = q_{c_A} = 2$ and $\mathcal{B} = 4$, where $\rho(i_1) = \{o, c_A\}$ and $[\underline{b}_{i_1}, \bar{b}_{i_1}] = [2, 2]$. Take $x_{i_1, o} = 2$ and $x_{i_1, c_A} = 0$. Grant i_1 is funded to its ceiling, so nothing can absorb the reserve standing in c_A and Axiom A holds; there is no unfunded grant and none funded below its ceiling, so Axiom B holds, and compliance is immediate. But $x_{i_1, c_A} = 0 < 2 = x_{i_1, o}$ while c_A retains its entire reserve, so Axiom E fails. $\square$

Proof of Proposition 2. Let $\mathcal{G}$ be an earmark problem, and let $\mathcal{G}'$ be an earmark problem which is equal to $\mathcal{G}$, except i submits at least one worse attribute (lower merit, higher cost $\underline{b}_i$ or $\bar{b}_i$, or fewer categories). Let $x = C^{\text{SR}}(\mathcal{G})$ and let $x' \equiv C^{\text{SR}}(\mathcal{G}')$.

⁴A script checking the four axioms against these four allocations directly, as they are stated in Sections 4.3 and 4.6, is available at <https://github.com/joejourden/GEB-R-R-R2/tree/main/code>.

We will show that if $x'_i > 0$ then $x_i > 0$; equivalently, if $x_i = 0$ then $x'_i = 0$. Use x^s to denote the intermediate step s of x , and x'^s to denote the intermediate step s of x' . Let $\rho(i)$ and $\rho'(i)$ denote the categories reported by i under $\mathcal{G}$ and $\mathcal{G}'$, respectively, and let $\underline{b}_i$ and $\underline{b}'_i$ denote lower bound costs under $\mathcal{G}$ and $\mathcal{G}'$, respectively.

Assume without loss of generality that in $\mathcal{G}'$, $i = i_{t'}$, and in $\mathcal{G}$, $i = i_t$, $t \leq t'$. The deviation cannot affect steps prior to t as only lower index grants i_s factor into equations (1) and (2), so $x^{t-1} = x'^{t-1} \leq x'^{t'-1}$, where the second inequality comes from the fact that x'^s increases in funding each step.

Suppose $x'_i > 0$. Since $\rho'(i) \subseteq \rho(i)$, and for each c , $x_c^{t-1} \leq x_c'^{t'-1}$, it follows

$$\begin{aligned} \sum_{c \in \rho(i)} (q_c - x_c^{t-1}) &\geq \sum_{c \in \rho'(i)} (q_c - x_c'^{t'-1}) \\ &\geq \underline{b}'_i \quad (\text{by Lemma 1}) \\ &\geq \underline{b}_i \quad (\text{by assumption.}) \end{aligned}$$

Therefore, by Lemma 1, $x_i > 0$. □

Proof of Proposition 3. Write $i_1 \succ \dots \succ i_n$ for the grants of I in merit order, let x^t denote the intermediate matrix after step t of the algorithm on $\mathcal{G}$, and let s be the position at which j is considered in $\mathcal{G}^{+j}$, so that j is preceded by $i_1, \dots, i_{s-1}$ and followed by $i_s, \dots, i_n$.

Steps $1, \dots, s-1$ are identical under $\mathcal{G}$ and $\mathcal{G}^{+j}$, since step t depends only on x^{t-1} and on the report of i_t , and neither is altered by adjoining a lower-indexed grant. In particular the column sums agree: x_c^{s-1} is the same under both problems for every c .

By hypothesis every funded grant outranks j , so $x_i = 0$ for each $i \in \{i_s, \dots, i_n\}$; that is, no grant considered after j receives funding under $\mathcal{G}$. Consequently no funding is assigned after step $s-1$, and $x_c = x_c^{s-1}$ for every c : the residual reserves facing j at its own step are exactly the residual reserves of the final allocation.

Now suppose $\underline{b}_j > \sum_{c \in \rho(j)} (q_c - x_c)$. By the previous paragraph this is precisely the condition $\sum_{c \in \rho(j)} (q_c - x_c^{s-1}) < \underline{b}_j$ under which the algorithm sets j 's row to zero, so j receives no funding and $x^s = x^{s-1}$. The remaining steps then face the same column sums as under $\mathcal{G}$ and, by Lemma 1, likewise assign nothing. Hence $C^{\text{SR}}(\mathcal{G}^{+j}) = (x, 0)$.

Without the affordability hypothesis, j may receive funding at step s , but only from reserves that no grant of $\{i_s, \dots, i_n\}$ draws upon, since those grants receive nothing under $\mathcal{G}$ and, facing weakly smaller residuals under $\mathcal{G}^{+j}$, receive nothing there either by Lemma 1. Grants $i_1, \dots, i_{s-1}$ were settled before step s . So $C^{\text{SR}}(\mathcal{G}^{+j})_i = x_i$ for every $i \in I$ in either case. □

Proof of Proposition 4. At step t the algorithm modifies row i_t only, and $i_1, \dots, i_n$ are distinct, so it halts after $n = |I|$ steps.

Fix a step t and write $k = |\rho(i_t)| \leq |\mathcal{C}|$ and $r_c = q_c - x_c^{t-1} \geq 0$ for the residual reserve of $c \in \rho(i_t)$. Computing $(r_c)_c$ and $R = \sum_{c \in \rho(i_t)} r_c$ costs $O(k)$, as does setting $x_{i_t}^t = 0$ when $R < \underline{b}_{i_t}$. Otherwise the step must solve (1)–(2) with $T := \min\{\bar{b}_{i_t}, R\} \leq R$.

Sort the residuals as $r_{(1)} \leq \dots \leq r_{(k)}$, and let $P_j = \sum_{l \leq j} r_{(l)}$, with $P_0 = 0$. For $j = 1, \dots, k$ let

$$A_j = P_j + (k - j) r_{(j)}$$

be the total that would be handed out if every category received $\min\{r_{(j)}, r_c\}$, i.e. if all categories were funded at the common level $r_{(j)}$ except those whose residual is smaller. Since

$$A_j - A_{j-1} = (k - j + 1)(r_{(j)} - r_{(j-1)}) \geq 0,$$

the sequence $A_1, \dots, A_k$ is nondecreasing, and $A_k = P_k = R \geq T$, so

$$j^* = \min\{j : A_j \geq T\} \quad \text{and} \quad \lambda = \frac{T - P_{j^*-1}}{k - j^* + 1}$$

are well defined. We claim $x_{i_t, c}^t = \min\{\lambda, r_c\}$ is the unique solution of (1)–(2).

First, $r_{(j^*-1)} < \lambda \leq r_{(j^*)}$: rearranging $A_{j^*} \geq T$ gives $r_{(j^*)} \geq \lambda$, and rearranging $A_{j^*-1} < T$, which holds by minimality of j^* , gives $r_{(j^*-1)} < \lambda$ (for $j^* = 1$, note $\lambda = T/k > 0$). Hence the $j^* - 1$ categories with the smallest residuals receive r_c and the remaining $k - j^* + 1$ receive λ , so $\sum_c x_{i_t, c}^t = P_{j^*-1} + (k - j^* + 1)\lambda = T$, which is (1). Since every category receives either λ or an amount at most $r_{(j^*-1)} < \lambda$, we have $\max_{c'} x_{i_t, c'}^t = \lambda$ and therefore $x_{i_t, c}^t = \min\{r_c, \max_{c'} x_{i_t, c'}^t\}$, which is (2).

For uniqueness, let y also solve (1)–(2) and put $\mu = \max_{c'} y_{c'}$, so that (2) gives $y_c = \min\{r_c, \mu\}$ for every c . If $\mu < \lambda$ then $\min\{r_c, \mu\} \leq \min\{r_c, \lambda\}$ for every c , with strict inequality for any c having $r_c > \mu$; at least one such category exists, since $\lambda \leq r_{(j^*)}$, and so $\sum_c y_c < T$, contradicting (1). The case $\mu > \lambda$ is symmetric. Hence $\mu = \lambda$ and $y = x_{i_t}^t$.

The computation is therefore: sort the k residuals, at cost $O(k \log k)$; form the prefix sums P_j , scan for j^* , evaluate λ , and set $x_{i_t, c}^t = \min\{\lambda, r_c\}$, each at cost $O(k)$; and update the column sums, at cost $O(k)$. Summing over the $|I|$ steps and using $k \leq |\mathcal{C}|$ gives the bound $O(|I| \cdot |\mathcal{C}| \cdot \log |\mathcal{C}|)$. $\square$

A.2 Floors

Two lemmas will aid in the proof of Theorem 2. The first shows that expanding an allocation which satisfies Axiom C cannot lead to an allocation failing to satisfy Axiom C. The second shows that procedures of the kind C_Δ^f runs in its floor phase terminate at attainable allocations that satisfy feasible floors. Its proof charges each increment of funding to a single category and tracks these charges cumulatively, which

yields both the budget bound and the slack that keeps each selected grant inside the budget set.

Lemma 2. *If ξ^A and ξ^B are two allocations, where $\xi^A \leq \xi^B$ and ξ^A satisfies Axiom C, then so does ξ^B .*

Proof of Lemma 2. Suppose $\xi^A \leq \xi^B$ and ξ^A satisfies Axiom C. For any category $c \in \mathcal{C}$, either:

1. $\xi_i^A = \bar{b}_i$ for all $i \in I_c$, so $\xi_i^B = \bar{b}_i$ for all $i \in I_c$, and ξ^B satisfies Axiom C for c ,
2. The highest-merit grant i^* satisfying $\xi_{i^*}^A < \bar{b}_{i^*}$ has $\xi_{i^*}^A = 0$, and $\sum_{i \in I_c} \xi_i^A + \underline{b}_{i^*} > q_c$. There are two sub-cases:
 - If $\xi_{i^*}^B > 0$, then $\sum_{i \in I_c} \xi_i^B \geq \sum_{i \in I_c} \xi_i^A + \underline{b}_{i^*} > q_c$. Hence, ξ^B satisfies Axiom C for c , or
 - If $\xi_{i^*}^B = 0$, then it remains that $\sum_{i \in I_c} \xi_i^A + \underline{b}_{i^*} > q_c$, and i^* is still the highest-merit grant satisfying $\xi_{i^*}^B < \bar{b}_{i^*}$. Hence, ξ^B satisfies Axiom C for c .
3. The highest-merit grant i^* satisfying $\xi_{i^*}^A < \bar{b}_{i^*}$ has $\underline{b}_{i^*} \leq \xi_{i^*}^A < \bar{b}_{i^*}$, but $\sum_{i \in I_c} \xi_i^A \geq q_c$. It follows that $\sum_{i \in I_c} \xi_i^B \geq q_c$, which implies ξ^B satisfies Axiom C for c .

□

Lemma 3. *Let $\xi^0 = 0$. At each step $t \geq 1$, if there are categories c such that ξ^{t-1} does not satisfy Axiom C for c , let $\mathcal{C}^t$ be a nonempty subset of them; otherwise, $\mathcal{C}^t = \emptyset$. If $\mathcal{C}^t$ is empty, let $\xi^t = \xi^{t-1}$, and the procedure ends. Otherwise, choose a grant i_t and a category $c^t \in \mathcal{C}^t \cap \rho(i_t)$ such that*

1. $\xi_{i_t}^{t-1} < \bar{b}_{i_t}$,
2. $\sum_{i \in I_{c^t}} \xi_i^{t-1} < q_{c^t}$, and
3. if $\xi_{i_t}^{t-1} = 0$ then also $\sum_{i \in I_{c^t} \setminus \{i_t\}} \xi_i^{t-1} + \underline{b}_{i_t} \leq q_{c^t}$,

and let

$$\xi_{i_t}^t = \xi_{i_t}^{t-1} + \min \left\{ \bar{b}_{i_t} - \xi_{i_t}^{t-1}, \max_{c \in \mathcal{C}^t \cap \rho(i_t)} d_c(\xi^{t-1}) \right\},$$

with $\xi_i^t = \xi_i^{t-1}$ for all $i \neq i_t$. (Such a pair exists: take any $c \in \mathcal{C}^t$ and let i_t be the highest-merit element of $A_c(\xi^{t-1})$, with $c^t = c$. Since ξ^{t-1} fails Axiom C at c , $d_c(\xi^{t-1}) > 0$ and $A_c(\xi^{t-1}) \neq \emptyset$, which gives (1) and (2); and if $\xi_{i_t}^{t-1} = 0$, failure of

Axiom C at c means $\underline{b}_{i_t} \leq d_c(\xi^{t-1})$, which is (3).) Any procedure of this type ends at a step $T \leq |I| + |\mathcal{C}| + 1$. Moreover, for every $t \leq T$, ξ^t is attainable and

$$\sum_{i \in I} \xi_i^t + d_c(\xi^t) \leq \sum_{c' \in \mathcal{C}} q_{c'} \leq \mathcal{B} \quad \text{for every } c \in \mathcal{C}, \quad (6)$$

and ξ^T satisfies Axiom C.

Proof of Lemma 3. Write $S_c(\xi) = \sum_{i \in I_c} \xi_i$, so that $d_c(\xi) = q_c - S_c(\xi)$. For each non-terminal step t , let $\gamma^t = \xi_{i_t}^t - \xi_{i_t}^{t-1}$ be the increment, and fix a category

$$\hat{c}^t \in \operatorname{argmax}_{c \in \mathcal{C}^t \cap \rho(i_t)} d_c(\xi^{t-1}), \quad \text{so that} \quad \gamma^t = \min \{ \bar{b}_{i_t} - \xi_{i_t}^{t-1}, d_{\hat{c}^t}(\xi^{t-1}) \}.$$

Note $i_t \in I_{\hat{c}^t}$, since $\hat{c}^t \in \rho(i_t)$.

Step 1: one-step bounds. Fix a non-terminal step t .

- (a) $\gamma^t \leq d_{\hat{c}^t}(\xi^{t-1})$ and $\xi_{i_t}^t \leq \bar{b}_{i_t}$, directly from the definition.
- (b) $\gamma^t > 0$: the first term is positive by (1), and $d_{\hat{c}^t}(\xi^{t-1}) \geq d_{c^t}(\xi^{t-1}) > 0$ by maximality and (2).
- (c) If $\xi_{i_t}^{t-1} = 0$, then $\gamma^t \geq \underline{b}_{i_t}$: the first term is $\bar{b}_{i_t} \geq \underline{b}_{i_t}$, and, because $\xi_{i_t}^{t-1} = 0$, condition (3) reads $\underline{b}_{i_t} \leq q_{c^t} - S_{c^t}(\xi^{t-1}) = d_{c^t}(\xi^{t-1}) \leq d_{\hat{c}^t}(\xi^{t-1})$.

Funding is nondecreasing along the run. By (c), a grant's first increment carries it to at least its lower bound, and by (a) no increment carries it past its upper bound. Hence for every t and every i , either $\xi_i^t = 0$ or $\underline{b}_i \leq \xi_i^t \leq \bar{b}_i$.

Step 2: cumulative charges. Charge the increment γ^t to the category $\hat{c}^t$, and let

$$\kappa_c^t = \sum_{s \leq t : \hat{c}^s = c} \gamma^s$$

be the total charged to c through step t , with $\kappa_c^0 = 0$. A grant can be selected at several steps; κ_c^t counts every increment charged to c , whichever grant received it and however many times that grant was funded. Three facts follow.

- (i) $\sum_{i \in I} \xi_i^t = \sum_{c \in \mathcal{C}} \kappa_c^t$. Each non-terminal step changes one coordinate, by γ^t , and charges γ^t to exactly one category; and $\xi^0 = 0$.
- (ii) $\kappa_c^t \leq S_c(\xi^t)$. Every increment charged to c was paid to a grant in I_c , so $\kappa_c^t \leq \sum_{s \leq t : i_s \in I_c} \gamma^s = S_c(\xi^t)$, all increments being nonnegative.

(iii) $\kappa_c^t \leq q_c$. By induction on t , the case $t = 0$ being trivial. If $\hat{c}^t \neq c$, then $\kappa_c^t = \kappa_c^{t-1} \leq q_c$. If $\hat{c}^t = c$, then by (ii) at $t - 1$ and Step 1(a),

$$\kappa_c^t = \kappa_c^{t-1} + \gamma^t \leq S_c(\xi^{t-1}) + d_c(\xi^{t-1}) = q_c.$$

Combining, for any t and any category c ,

$$\sum_{i \in I} \xi_i^t = \kappa_c^t + \sum_{c' \neq c} \kappa_{c'}^t \leq S_c(\xi^t) + \sum_{c' \neq c} q_{c'} = \sum_{c' \in \mathcal{C}} q_{c'} - d_c(\xi^t),$$

which is (6); the second inequality there is the standing assumption $\sum_c q_c \leq \mathcal{B}$ on floor problems. Directly from (i) and (iii), $\sum_i \xi_i^t = \sum_c \kappa_c^t \leq \sum_c q_c \leq \mathcal{B}$. Together with Step 1, ξ^t is attainable.

Step 3: termination and Axiom C. At each non-terminal step one of two things happens. If $\gamma^t = \bar{b}_{i_t} - \xi_{i_t}^{t-1}$, grant i_t becomes fully funded and, by (1), is never selected again. Otherwise $\gamma^t = d_{\hat{c}^t}(\xi^{t-1})$, so $d_{\hat{c}^t}(\xi^t) = 0$; as funding is nondecreasing, $d_{\hat{c}^t} \leq 0$ at all later steps, so $\hat{c}^t$ satisfies Axiom C from then on and never again belongs to $\mathcal{C}^s$, and in particular is never again charged. Hence there are at most $|I| + |\mathcal{C}|$ non-terminal steps, and $T \leq |I| + |\mathcal{C}| + 1$. The procedure ends only when $\mathcal{C}^T = \emptyset$, that is, when $\xi^{T-1} = \xi^T$ satisfies Axiom C for every category. $\square$

Proof of Theorem 2. To show the function is a choice rule, it suffices to show that the procedure terminates in finite time and it has attainable outcomes.

Finite termination, attainable outcomes, and Axiom C (satisfying feasible floors). Call the steps of C_Δ^f with $\mathcal{C}^t \neq \emptyset$ the *floor phase* and the remaining non-terminal steps the *open phase*.

The floor phase is a run of the procedure in Lemma 3, and it never stalls on the budget. Suppose ξ^{t-1} was reached by floor-phase steps that are steps of that procedure, and $\mathcal{C}^t \neq \emptyset$. The grant i_t chosen by C_Δ^f is the highest-merit element of $\bigcup_{c \in \mathcal{C}^t} A_c(\xi^{t-1})$, so $i_t \in A_c(\xi^{t-1})$ for some $c \in \mathcal{C}^t$; fix such a c and set $c^t = c$. Then i_t is also the highest-merit element of $A_{c^t}(\xi^{t-1})$, so conditions (1)–(3) of Lemma 3 hold by the existence argument given there, and the increment prescribed by C_Δ^f , $\min\{\bar{b}_{i_t} - \xi_{i_t}^{t-1}, M_t\}$ with $M_t = \max_{c \in \mathcal{C}^t \cap \rho(i_t)} d_c(\xi^{t-1})$, is exactly the one in Lemma 3. It remains to check that C_Δ^f does not halt here, i.e. that the ξ^{t-1} -budget set is nonempty; we show $i_t \in \mathcal{B}_{\xi^{t-1}}$. Apply (6) at $t - 1$ to the category c^t , which has $d_{c^t}(\xi^{t-1}) > 0$ by (2):

$$\sum_{i \in I} \xi_i^{t-1} + d_{c^t}(\xi^{t-1}) \leq \mathcal{B}.$$

If $\xi_{i_t}^{t-1} = 0$, condition (3) gives $\underline{b}_{i_t} \leq d_{c^t}(\xi^{t-1})$, so $\sum_i \xi_i^{t-1} + \underline{b}_{i_t} \leq \mathcal{B}$. If $\xi_{i_t}^{t-1} > 0$, attainability of ξ^{t-1} and (1) give $\underline{b}_{i_t} \leq \xi_{i_t}^{t-1} < \bar{b}_{i_t}$, and $\sum_i \xi_i^{t-1} \leq \mathcal{B} - d_{c^t}(\xi^{t-1}) < \mathcal{B}$.

In either case $i_t \in \mathcal{B}_{\xi^{t-1}}$. The selected increment itself also fits the remaining budget: $\sum_i \xi_i^t \leq \mathcal{B}$ by (6) at t . By induction, the floor phase of C_Δ^f coincides with a run of the procedure in Lemma 3, never meets an empty budget set, and ends after some step $l \leq |I| + |\mathcal{C}|$ at an attainable ξ^l that satisfies Axiom C.

The open phase. Since $\xi^{t-1} \geq \xi^l$ for every $t > l$, Lemma 2 gives $\mathcal{C}^t = \emptyset$ at every later step, so once the open phase begins it never returns to the floor phase, and Axiom C holds at every later allocation. At an open-phase step, $i_t \in \mathcal{B}_{\xi^{t-1}}$ and $M_t = \mathcal{B} - \sum_i \xi_i^{t-1}$, so the total after the step is at most $\mathcal{B}$ and $\xi_{i_t}^t \leq \bar{b}_{i_t}$. If $\xi_{i_t}^{t-1} = 0$, membership in the budget set gives $\underline{b}_{i_t} \leq M_t$, and $\underline{b}_{i_t} \leq \bar{b}_{i_t}$, so $\xi_{i_t}^t \geq \underline{b}_{i_t}$. Hence attainability is preserved. Each open-phase step either fully funds i_t or brings the total to $\mathcal{B}$; in the latter case the budget set is empty, because $\underline{b}_i > 0$ for every i , and C_Δ^f halts at the next step. So the open phase has finitely many steps, and C_Δ^f terminates at an attainable allocation satisfying Axiom C.

Axiom A (non-wastefulness). By construction the algorithm cannot terminate at any step t if there is a grant in the ξ^{t-1} -budget set. It follows that the choice rule is non-wasteful.

Axiom B (respecting priorities). Let $\xi = C_\Delta^f(\mathcal{G})$. Fix some $j \in I$ with $\underline{b}_j \leq \xi_j < \bar{b}_j$. Assume for contradiction that there is some $i \in I$ such that $j \succ i$, $\rho(j) \supseteq \rho(i)$, and $\xi_i > 0$. Then there was some step t when i was funded.

- If $\rho(j) \cap \mathcal{C}^t \neq \emptyset$, then $\xi_j < \bar{b}_j$ during step t . Hence, i is the highest-merit grant in $\bigcup_{c \in \mathcal{C}^t} I_c$ with funding less than its maximum amount, a contradiction since $j \in \bigcup_{c \in \mathcal{C}^t} I_c$ has $j \succ i$ and $\underline{b}_j \leq \xi_j < \bar{b}_j$.
- Now assume $\rho(j) \cap \mathcal{C}^t = \emptyset$. Therefore, i is the highest-merit grant in the ξ^{t-1} -budget set, a contradiction since $j \succ i$ and $\underline{b}_j \leq \xi_j < \bar{b}_j$.

This establishes the second clause of Axiom B. We now turn to the first.

Again, let $\xi = C_\Delta^f(\mathcal{G})$. Fix some $j \in I$ with $\xi_j = 0$. Suppose there is some subset $\{h_1, \dots, h_n\} \subseteq I$ such that (1) $\xi_{h_k} > 0$, $j \succ h_k$, and $\rho(j) \supseteq \rho(h_k)$ for all k , and (2) $\sum_{k=1}^n \xi_{h_k} \geq \underline{b}_j$. Let t be the first step of C_Δ^f for which $\xi_{h_k}^t > 0$ for some h_k among $\{h_1, \dots, h_n\}$. Let l be the last step for which $\mathcal{C}^l \neq \emptyset$.

Suppose $t \leq l$. Let i_t be the grant funded during step t , which implies $i_t \in \{h_1, \dots, h_n\}$. Then i_t was the highest-merit grant among $\bigcup_{c \in \mathcal{C}^t} I_c$ having $\xi_{i_t} < \bar{b}_{i_t}$. Since i_t was funded at this step, it follows that there is no $\tilde{i} \in I$ having $\xi_{\tilde{i}}^t < \bar{b}_{\tilde{i}}$ and satisfying $\rho(\tilde{i}) \supseteq \rho(i_t)$ and $\tilde{i} \succ i_t$, but this contradicts j .

Now suppose $t > l$, which implies no grant among $\{h_1, \dots, h_n\}$ received any funding by the end of step l . Then, at the start of the earliest step t for which one of $\{h_1, \dots, h_n\}$ received funding, the ξ^{t-1} -budget set includes all grants with minimum cost less than or equal to $\sum_{k=1}^n \xi_{h_k}$. Therefore, as one of $h_1, \dots, h_n$ was the highest-

merit grant in the budget set, there is no $\tilde{i} \in I$ having $\xi_{\tilde{i}}^t < \bar{b}_{\tilde{i}}$, and satisfying both $\underline{b}_{\tilde{i}} \leq \sum_{k=1}^n \xi_{h_k}$ and $\tilde{i} \succ h_k$ for all k , but this contradicts j .

Monotonicity in cost. Let $\xi = C_{\Delta}^f(\mathcal{G})$, and fix a grant $i^* \in I$. Fix a deviation consisting of an increase in cost for i^* . Let $\mathcal{G}'$ be equal to $\mathcal{G}$ except having this deviation, and let $\xi' = C_{\Delta}^f(\mathcal{G}')$.

Claim 2.1. (No gain from disimprovement.) *If $\xi_{i^*} = 0$, then $\xi'_{i^*} = 0$.*

Proof. Write $\underline{b}'_{i^*} \geq \underline{b}_{i^*}$ and $\bar{b}'_{i^*} \geq \bar{b}_{i^*}$ for the cost bounds of i^* in $\mathcal{G}'$; every other component of the two problems coincides, including the merit order. Assume $\xi_{i^*} = 0$. We show by induction that the two runs coincide step by step, which gives $\xi' = \xi$ and in particular $\xi'_{i^*} = 0$. We first record three facts about the run of C_{Δ}^f on $\mathcal{G}$.

(P1) *Every non-terminal step assigns a strictly positive amount.* The increment is $\min\{\bar{b}_{i_t} - \xi_{i_t}^{t-1}, M_t\}$. The first term is positive because i_t is selected from $\bigcup_{c \in \mathcal{C}^t} A_c(\xi^{t-1})$ or from the budget set, both of which contain only grants below their ceilings. If $\mathcal{C}^t \neq \emptyset$, then $M_t \geq d_c(\xi^{t-1}) > 0$ for any $c \in \mathcal{C}^t \cap \rho(i_t)$, since Axiom C can fail at c only when $d_c(\xi^{t-1}) > 0$. If $\mathcal{C}^t = \emptyset$, then i_t lies in the ξ^{t-1} -budget set, so $M_t = \mathcal{B} - \sum_i \xi_i^{t-1} > 0$.

(P2) *A grant's first funding is at least its lower bound.* Suppose $\xi_{i_t}^{t-1} = 0$. If $\mathcal{C}^t \neq \emptyset$, then i_t is the highest-merit element of $\bigcup_{c \in \mathcal{C}^t} A_c(\xi^{t-1})$, so for any $c \in \mathcal{C}^t \cap \rho(i_t)$ it is the highest-merit element of $A_c(\xi^{t-1})$. Axiom C fails at c and $\xi_{i_t}^{t-1} = 0$, so $\underline{b}_{i_t} \leq d_c(\xi^{t-1}) \leq M_t$. If $\mathcal{C}^t = \emptyset$, membership of i_t in the budget set gives $\underline{b}_{i_t} \leq \mathcal{B} - \sum_i \xi_i^{t-1} = M_t$. In either case the increment $\min\{\bar{b}_{i_t}, M_t\}$ is at least $\underline{b}_{i_t}$. Since funding never falls, every grant funded at some step t satisfies $\xi_i^s \geq \underline{b}_i$ at all later steps $s \geq t$.

(P3) *At every non-terminal step t , $i_t \neq i^*$ and i_t belongs to the ξ^{t-1} -budget set.* By (P1), $\xi_{i_t} > 0 = \xi_{i^*}$, so $i_t \neq i^*$. If $\xi_{i_t}^{t-1} = 0$, then by (P2) and the attainability of ξ (established above in the proof of finite termination and attainability), $\sum_i \xi_i^{t-1} + \underline{b}_{i_t} \leq \sum_i \xi_i^t \leq \sum_i \xi_i \leq \mathcal{B}$. If $\xi_{i_t}^{t-1} > 0$, then $\underline{b}_{i_t} \leq \xi_{i_t}^{t-1} < \bar{b}_{i_t}$ by (P2) and the selection rule, and $\sum_i \xi_i^{t-1} < \sum_i \xi_i^t \leq \mathcal{B}$ by (P1). Either way $i_t \in \mathcal{B}_{\xi^{t-1}}$.

Now fix a step t of $C_{\Delta}^f(\mathcal{G})$ and assume $\xi^{t-1} = \xi'^{t-1} =: \eta$. Since funding never falls and $\xi_{i^*} = 0$, $\eta_{i^*} = 0$. Write $\mathcal{B}_{\eta}$ and $\mathcal{B}'_{\eta}$ for the η -budget sets in $\mathcal{G}$ and $\mathcal{G}'$, and U and U' for the sets of categories at which η fails Axiom C in $\mathcal{G}$ and $\mathcal{G}'$.

Budget sets. Membership of any $i \neq i^*$ depends only on η and i 's own data, so it is the same in both problems. Since $\eta_{i^*} = 0$, $i^* \in \mathcal{B}'_{\eta}$ means $\sum_i \eta_i + \underline{b}'_{i^*} \leq \mathcal{B}$, which implies $\sum_i \eta_i + \underline{b}_{i^*} \leq \mathcal{B}$. Hence $\mathcal{B}'_{\eta} \subseteq \mathcal{B}_{\eta}$, and the two sets differ at most in i^* .

Axiom C. Fix a category c . The shortfall $d_c(\eta)$ depends only on η and on eligibility, so it is the same in both problems; if $d_c(\eta) \leq 0$, Axiom C holds at c in both. Suppose $d_c(\eta) > 0$. The set $A_c(\eta)$ is also the same in both problems: membership of $i \neq i^*$ is unchanged, and if $c \in \rho(i^*)$ then $\eta_{i^*} = 0 < \bar{b}_{i^*} \leq \bar{b}'_{i^*}$, so i^* belongs to $A_c(\eta)$ in both. As merit is unchanged, $A_c(\eta)$ has the same highest-merit element i° in both

problems. If $i^\circ \neq i^*$, the test of Axiom C at c involves only η_{i° , $\underline{b}_{i^\circ}$ and $d_c(\eta)$, and returns the same verdict in both problems. If $i^\circ = i^*$, the test reads $\underline{b}_{i^*} > d_c(\eta)$ in $\mathcal{G}$ and $\underline{b}'_{i^*} > d_c(\eta)$ in $\mathcal{G}'$, and the first implies the second. Therefore

$$U' \subseteq U, \quad \text{and} \quad c \in U \setminus U' \text{ only if the highest-merit element of } A_c(\eta) \text{ is } i^*. \quad (7)$$

Termination. If step t is terminal for $\mathcal{G}$, then $\mathcal{B}_\eta = \emptyset$, so $\mathcal{B}'_\eta = \emptyset$ and the run on $\mathcal{G}'$ terminates at the same step with the same vector η . Otherwise, by (P3), $i_t \in \mathcal{B}_\eta$ and $i_t \neq i^*$, so $i_t \in \mathcal{B}'_\eta$ and step t is non-terminal for $\mathcal{G}'$ as well. It remains to show that the same grant receives the same amount.

Case $U = \emptyset$. By (7), $U' = \emptyset$, so $\mathcal{C}^t = \mathcal{C}'^t = \emptyset$ and both runs select the highest-merit element of their budget set. Since i_t is the highest-merit element of $\mathcal{B}_\eta$ and belongs to the subset $\mathcal{B}'_\eta$, it is also the highest-merit element of $\mathcal{B}'_\eta$. The amount $M_t = \mathcal{B} - \sum_i \eta_i$ is the same in both problems, and $\bar{b}_{i_t}$ is unchanged because $i_t \neq i^*$.

Case $U \neq \emptyset$. Then $\mathcal{C}^t \neq \emptyset$ and i_t is the highest-merit element of $\bigcup_{c \in \mathcal{C}^t} A_c(\eta)$. In particular $\mathcal{C}^t \cap \rho(i_t) \neq \emptyset$, and for every $c \in \mathcal{C}^t \cap \rho(i_t)$ the highest-merit element of $A_c(\eta)$ is $i_t \neq i^*$. (This is the sense in which i_t outranks i^* : if i^* belongs to $\bigcup_{c \in \mathcal{C}^t} A_c(\eta)$, then $i_t \succ i^*$, and in any case i^* is not the next-best grant of any category in $\mathcal{C}^t \cap \rho(i_t)$.) We show:

- (a) $\rho(i_t) \cap \mathcal{C}'^t = \rho(i_t) \cap \mathcal{C}^t$, and
- (b) $\mathcal{C}'^t \subseteq \mathcal{C}^t$.

By (7), every $c \in \mathcal{C}^t \cap \rho(i_t)$ lies in U' . Groups of Δ with smaller index than δ_t contain no category of U , hence none of U' , while δ_t contains a category of $\mathcal{C}^t \cap \rho(i_t) \subseteq U'$. So the active group in $\mathcal{G}'$ is also δ_t , and $\mathcal{C}'^t = U' \cap \delta_t \subseteq U \cap \delta_t = \mathcal{C}^t$, which is (b). Since $\mathcal{C}^t \cap \rho(i_t) \subseteq U' \cap \delta_t = \mathcal{C}'^t$, (b) gives (a).

The sets $A_c(\eta)$ coincide in the two problems, so by (b), $\bigcup_{c \in \mathcal{C}'^t} A_c(\eta) \subseteq \bigcup_{c \in \mathcal{C}^t} A_c(\eta)$, and by (a) the smaller union still contains i_t . Being the highest-merit element of the larger union, i_t is the highest-merit element of the smaller one, so $i'_t = i_t$. By (a), and since $i_t \neq i^*$,

$$\xi_{i_t}^t = \eta_{i_t} + \min \left\{ \bar{b}_{i_t} - \eta_{i_t}, \max_{c \in \mathcal{C}^t \cap \rho(i_t)} d_c(\eta) \right\} = \eta_{i_t} + \min \left\{ \bar{b}_{i_t} - \eta_{i_t}, \max_{c \in \mathcal{C}'^t \cap \rho(i_t)} d_c(\eta) \right\} = \xi_{i_t}'^t.$$

In both cases the same grant i_t receives the same amount and no other grant's funding changes, so $\xi'^t = \xi^t$. By induction the two runs coincide and terminate together, so $\xi' = \xi$, and $\xi'_{i^*} = \xi_{i^*} = 0$. $\square$

Claim 2.2. (No harm from improvement.) *If $\xi'_{i^*} > 0$, then $\xi_{i^*} > 0$.*

Proof. This follows directly from Claim 2.1. Since $\xi'_{i^*} = 0$ whenever $\xi_{i^*} = 0$, the contrapositive holds: $\neg(\xi_{i^*} = 0)$ whenever $\neg(\xi'_{i^*} = 0)$. But this is equivalent to Claim 2.2, $\xi_{i^*} > 0$ whenever $\xi'_{i^*} > 0$. $\square$

Failure of monotonicity in merit and categories. See Example 6.

Independence of non-contending grants. Consider a floor problem $\mathcal{G}$ with grants I , and a floor problem $\mathcal{G}^{+j}$ with grants $I \cup \{j\}$, which is the same but adds a new grant j which is non-contending at ξ . In the floor phase of C_Δ^f , since the grant j is non-marginal, it does not affect whether feasible floors are satisfied for any category. Moreover, in each of its categories there is always a grant with higher merit that is not fully funded, so the stopping rule for consideration of each category dictates that j is never funded in this phase. We argue by induction on the steps of the algorithm that the runs on $\mathcal{G}$ and $\mathcal{G}^{+j}$ produce the same sequence $\xi^0, \xi^1, \dots$, with $\xi_j^t = 0$ throughout. Suppose they agree through step $t - 1$. Since funding only increases along the algorithm, the second condition of non-marginality at the final allocation ξ holds a fortiori at ξ^{t-1} : in every $c \in \rho(j)$ some $i \succ j$ has $\xi_i^{t-1} < \bar{b}_i$. Hence j is never the highest-merit grant in $A_c(\xi^{t-1})$, so adding j alters neither which categories satisfy Axiom C, and thus $\mathcal{C}^t$, nor the grant i_t chosen when $\mathcal{C}^t \neq \emptyset$. If $\mathcal{C}^t = \emptyset$ and $\mathcal{B}_{\xi^{t-1}}$ is nonempty in $\mathcal{G}$, the grant i_t selected there receives positive funding, since $\underline{b}_{i_t} > 0$; so $\xi_{i_t} > 0$ and, j being non-contending, $i_t \succ j$. The budget set in $\mathcal{G}^{+j}$ differs from that in $\mathcal{G}$ at most by j , so the same i_t is selected. Finally, when the run on $\mathcal{G}$ terminates at ξ , its budget set is empty, and by the first condition of non-marginality $\underline{b}_j > \mathcal{B} - \sum_{i \in I} \xi_i$, so $j \notin \mathcal{B}_\xi$ in $\mathcal{G}^{+j}$ either. Both runs therefore terminate at the same step, and $C_\Delta^f(\mathcal{G}^{+j})$ coincides with $C_\Delta^f(\mathcal{G})$ on I and assigns j nothing. $\square$

Proof of Proposition 5. Fix an ordered partition Δ , a floor problem $\mathcal{G}$, and a grant $j \in I$ that is non-marginal at $\xi := C_\Delta^f(\mathcal{G})$. Let $\theta_j \succeq_t \theta'_j$ and let $\mathcal{G}'$ be $\mathcal{G}$ with j 's report replaced by θ'_j .

Non-marginality is defined only at grants with $\xi_j = 0$, and since funding is non-decreasing along the run, $\xi_j^t = 0$ at every step. By the second condition of non-marginality, for each $c \in \rho(j)$ fix a witness $w_c \in I_c$ with $w_c \succ j$ and $\xi_{w_c} < \bar{b}_{w_c}$; again by monotonicity of the run, $\xi_{w_c}^t < \bar{b}_{w_c}$ at every step. Note that $\rho(w_c)$, m_{w_c} and the cost bounds of w_c are unchanged in $\mathcal{G}'$.

We show by induction that the two runs coincide. They agree at $\xi^0 = \xi'^0 = 0$. Suppose they agree through step $t - 1$, and write η for the common funding vector, so $\eta_j = 0$.

(a) The satisfied categories are the same. Fix c . If $c \notin \rho(j)$, nothing bearing on c differs between the problems, so assume $c \in \rho(j)$ and let $w = w_c$. Check Axiom C at c . The shortfall is the same under both problems: the only term of $\sum_{i \in I_c} \eta_i$

that could differ is $\eta_j = 0$, whether or not j has dropped c , so $d_c(\eta)$ is unchanged and in particular $d_c(\eta) > 0$ has the same truth value. Suppose $d_c(\eta) > 0$. Under $\mathcal{G}$ the set $A_c(\eta)$ contains w , since $w \in I_c$ and $\eta_w < \bar{b}_w$, and $w \succ j$; so the highest-merit element of $A_c(\eta)$ is not j . Passing to $\mathcal{G}'$ either deletes j from $A_c(\eta)$ or lowers its merit, and changes nothing else, so that element and its lower bound are unchanged. Hence Axiom C holds at c under $\mathcal{G}$ if and only if it holds under $\mathcal{G}'$, and $\mathcal{C}^t$ and δ_t coincide.

(b) The same grant is funded, by the same amount. Suppose $\mathcal{C}^t \neq \emptyset$. The algorithm selects the highest-merit element of $\{i \in \bigcup_{c \in \mathcal{C}^t} I_c : \eta_i < \bar{b}_i\}$. If j belongs to this set under $\mathcal{G}$, then $c \in \mathcal{C}^t \cap \rho(j)$ for some c , and w_c lies in the same set with $w_c \succ j$; so $i_t \neq j$. Passing to $\mathcal{G}'$ can only delete j from the set or lower its merit, leaving every other element untouched, so i_t is unchanged. The amount M_t depends only on the floors, on the totals $\sum_{i \in I_c} \eta_i$, which are unchanged by (a), and on $\rho(i_t)$ with $i_t \neq j$, so the increment $\min\{\bar{b}_{i_t} - \eta_{i_t}, M_t\}$ is unchanged.

Suppose instead $\mathcal{C}^t = \emptyset$. The algorithm selects the highest-merit element of the η -budget set. Were that j , grant j would receive funding at step t , contradicting $\xi_j = 0$; so $i_t \neq j$. Under $\mathcal{G}'$ the budget set can differ only in whether it contains j , since $\underline{b}'_j \geq \underline{b}_j$ and no other grant's parameters change, and j was not the maximum in any case. So i_t is unchanged, as is $M_t = \mathcal{B} - \sum_i \eta_i$.

(c) The runs stop together. $C_\Delta^f(\mathcal{G})$ halts at the first step whose budget set is empty. At that step, membership under $\mathcal{G}'$ can only be more restrictive: for j , exclusion under $\mathcal{G}$ means $\sum_i \eta_i + \underline{b}_j > \mathcal{B}$, and $\underline{b}'_j \geq \underline{b}_j$; every other grant is unchanged. So $C_\Delta^f(\mathcal{G}')$ halts no later than $C_\Delta^f(\mathcal{G})$.

It remains to rule out an earlier halt under $\mathcal{G}'$. The two budget sets differ only in whether they contain j , so at a step where $\mathcal{G}$ does not halt, $\mathcal{G}'$ could halt only if $\mathcal{B}_\eta = \{j\}$. Suppose it were. Then $\sum_i \eta_i < \sum_i \eta_i + \underline{b}_j \leq \mathcal{B}$, so any $i \neq j$ with $\underline{b}_i \leq \eta_i < \bar{b}_i$ would itself lie in $\mathcal{B}_\eta$; hence every $i \neq j$ is either fully funded or has $\eta_i = 0$ and $\sum_k \eta_k + \underline{b}_i > \mathcal{B}$. If $\mathcal{C}^t = \emptyset$, the algorithm selects the highest-merit element of $\mathcal{B}_\eta = \{j\}$, giving $i_t = j$ and contradicting (b). If $\mathcal{C}^t \neq \emptyset$, then $i_t \neq j$ by (b) and $\eta_{i_t} < \bar{b}_{i_t}$, so $\eta_{i_t} = 0$ and $\sum_k \eta_k + \underline{b}_{i_t} > \mathcal{B}$. Grant i_t receives funding at step t , so $\xi_{i_t} > 0$ and therefore $\xi_{i_t} \geq \underline{b}_{i_t}$; since funding is nondecreasing along the run, $\sum_i \xi_i \geq \sum_i \eta_i + \underline{b}_{i_t} > \mathcal{B}$, contradicting the attainability of ξ established in the proof of Theorem 2. So $\mathcal{B}_\eta \neq \{j\}$ at every non-terminal step, and the two runs halt together with the same allocation. $\square$

Proof of Proposition 6. Step bound. Every non-terminal step increases the funding of exactly one grant by $\Delta\xi^t = \min\{\bar{b}_{i_t} - \xi_{i_t}^{t-1}, M_t\}$, and this quantity is strictly positive: the selection rule gives $\xi_{i_t}^{t-1} < \bar{b}_{i_t}$, and $M_t > 0$ because either $\mathcal{C}^t \cap \rho(i_t) \neq \emptyset$, in which case any such category is unsatisfied and hence has $\sum_{i \in I_c} \xi_i^{t-1} < q_c$ by Axiom C, or $\mathcal{C}^t = \emptyset$ and i_t lies in the ξ^{t-1} -budget set, so $\mathcal{B} - \sum_i \xi_i^{t-1} > 0$. Whichever term attains the minimum then determines which of three events occurs. If $\bar{b}_{i_t} - \xi_{i_t}^{t-1} \leq M_t$, grant

i_t becomes fully funded; since funding never falls and fully funded grants are excluded from later selection, this occurs at most $|I|$ times. If M_t attains the minimum and $\mathcal{C}^t \neq \emptyset$, let c^t attain the maximum defining M_t ; then $i_t \in I_{c^t}$ and $\sum_{i \in I_{c^t}} \xi_i^t = q_{c^t}$, so c^t becomes satisfied, and by Lemma 2 it remains so; this occurs at most $|\mathcal{C}|$ times, and it also implies that the index of the active group δ_t is nondecreasing. If M_t attains the minimum and $\mathcal{C}^t = \emptyset$, the budget is exhausted and the algorithm terminates; this occurs at most once. Hence $T \leq |I| + |\mathcal{C}| + 1$.

Per-step cost. The implementation stores ξ and the running totals $S_c = \sum_{i \in I_c} \xi_i$. Testing Axiom C at a category needs S_c and the highest-merit not-fully-funded grant of I_c , found by scanning the merit-ordered list I_c ; so constructing δ_t and $\mathcal{C}^t$ costs $O(\sum_c |I_c|) \subseteq O(|I| \cdot |\mathcal{C}|)$, and selecting i_t costs no more. Given the totals retained from that sweep, computing M_t and updating ξ_{i_t} and $(S_c)_{c \in \rho(i_t)}$ costs $O(|\mathcal{C}|)$. Multiplying $O(|I| \cdot |\mathcal{C}|)$ per step by the step bound gives the result. $\square$

A.3 Merit-Maximal Choice Rules

Proof of Theorem 3. The merit-maximal rule $C_{\mathcal{M}}^{\text{mm}}$ is uniquely determined, since $\Xi(\mathcal{G})$ is a finite non-empty totally ordered set and therefore has a single maximum. It suffices, then, to show that $C_{\mathcal{M}}^{\text{mm}}$ itself fails each of the listed properties: any $C \in \mathcal{M}$ possessing one of them differs from $C_{\mathcal{M}}^{\text{mm}}$ and so is not merit-maximal on $\mathcal{M}$. Independence of non-contending grants fails by Example 7, and monotonicity in merit and in categories by Example 6. It remains to treat monotonicity in cost, and monotonicity in types for non-marginal grants. To do this, we consider Example 5 and construct a monotonic disimprovement for a grant which is non-marginal.

The first step is finding the allocation $C_{\mathcal{M}}^{\text{mm}}(\mathcal{G}) = (4, 12, 0)$. We first check $(4, 12, 0) \in \Xi(\mathcal{G})$. It exhausts the budget, so it is non-wasteful (Axiom A). Its category total is $12 \geq q_c$, so $d_c(\xi) \leq 0$ and Axiom C is satisfied. For Axiom B, the grants funded short of their ceilings are i_1 and i_2 : i_1 has no lower-priority grant, since $\rho(i_1) = \emptyset \not\supseteq \{c\}$, and the only lower-priority grant of i_2 is i_3 , which is unfunded, so condition 2 holds. The only unfunded grant, i_3 , has no funded grant below it, so condition 1 holds as well.

No allocation with $\xi_{i_1} > 4$ lies in $\Xi(\mathcal{G})$. Suppose $\xi_{i_1} > 4$, so that $\xi_{i_2} + \xi_{i_3} < 12 = q_c$ and Axiom C must therefore bind. Since $\xi_{i_2} \leq \mathcal{B} - \xi_{i_1} < 12 < 13 = \bar{b}_2$, grant i_2 is not fully funded, so i_2 is the highest-merit member of $\{i \in I_c : \xi_i < \bar{b}_i\}$ and Axiom C requires $\xi_{i_2} = 0$ together with $\xi_{i_3} + \bar{b}_2 > 12$, i.e. $\xi_{i_3} > 5$. But $\bar{b}_3 = 4$, so this is impossible. (Concretely, the only non-wasteful allocation with $\xi_{i_1} = 10$ is $(10, 0, 4)$: at $(10, 0, 3)$ or below, i_3 can still absorb unassigned funds, while at $(10, 0, 4)$ the two unassigned units cannot be placed anywhere. And $(10, 0, 4)$ fails Axiom C, because $4 + \bar{b}_2 = 11 \leq 12$.) Among allocations with $\xi_{i_1} = 4$, the budget permits at most $\xi_{i_2} = 12$, and $(4, 12, 0)$ therefore dominates every other element of $\Xi(\mathcal{G})$.

We will show that i_3 is non-marginal at $C_{\mathcal{M}}^{\text{mm}}(\mathcal{G})$. Write $\xi = (4, 12, 0)$. Condition 1 holds: $\underline{b}_3 = 2 > 0 = \mathcal{B} - \sum_i \xi_i$. Condition 2 holds: $\rho(i_3) = \{c\}$, and i_2 satisfies $\rho(i_2) \supseteq \rho(i_3)$ and $i_2 \succ i_3$, hence $i_2 \succ_{\rho} i_3$, with $\xi_{i_2} = 12 < 13 = \bar{b}_2$. (Grant i_3 is moreover non-contending at ξ , since the funded grants i_1 and i_2 both outrank it; this is not needed here, but it is what the independence property would additionally require.) So i_3 sits at neither margin: the budget is exhausted and it cannot absorb the residual, and in its only category a strictly higher-priority grant is still short of its ceiling.

It remains to construct a monotonic deviation. Let $\mathcal{G}'$ be identical to $\mathcal{G}$ except that i_3 reports $[\underline{b}'_3, \bar{b}'_3] = [2, 6]$. Since $\underline{b}'_3 = \underline{b}_3$, $\bar{b}'_3 > \bar{b}_3$, the claimed categories are unchanged, and $m'_3 \leq m_3$, we have $\theta_{i_3} \succsim_t \theta'_{i_3}$: the deviation is a weak disimprovement. If $m'_3 = m_3$, the disimprovement is in cost alone.

Then $C_{\mathcal{M}}^{\text{mm}}(\mathcal{G}') = (10, 0, 6)$. This allocation exhausts the budget, so Axiom A holds. Its category total is $6 < q_c$, but i_3 is now fully funded at $\bar{b}'_3 = 6$, so i_2 is the highest-merit member of $\{i \in I_c : \xi_i < \bar{b}_i\}$, with $\xi_{i_2} = 0$ and $6 + \underline{b}_2 = 13 > 12$, so Axiom C is satisfied. For Axiom B, i_1 and i_3 are fully funded, so condition 2 of Axiom B is vacuous, and the only unfunded grant is i_2 , whose sole lower-priority funded grant contributes $\xi_{i_3} = 6 < 7 = \underline{b}_2$. Since $\bar{b}_1 = 10$, no allocation gives i_1 more, and $(10, 0, 6)$ is the only non-wasteful allocation with $\xi_{i_1} = 10$; hence it is the maximum of $\Xi(\mathcal{G}')$.

Therefore $C_{\mathcal{M}}^{\text{mm}}(\mathcal{G})_{i_3} = 0$ while $C_{\mathcal{M}}^{\text{mm}}(\mathcal{G}')_{i_3} = 6 > 0$, with i_3 non-marginal at $C_{\mathcal{M}}^{\text{mm}}(\mathcal{G})$. Monotonicity in types therefore fails, and under the reading in which $m'_3 = m_3$, so does monotonicity in cost. $\square$

A.4 Strong Feasible Floors

Recall Axiom C* from Section 4.4: whenever category c falls short of its floor, every grant eligible for c that could still absorb funding, that is, every grant in $A_c(\xi)$, is unfunded and has a minimum cost exceeding the shortfall $d_c(\xi)$, so that funding any of them would carry c past its floor. Since the binding case is the cheapest eligible grant that is not fully funded, C* demands more funding before a floor counts as met than Axiom C does.

Proposition 10. *If a choice rule C satisfies non-wastefulness (Axiom A), respects priorities (Axiom B) and satisfies strong feasible floors (Axiom C*), it is not independent of non-contending grants.*

Proof of Proposition 10. Suppose there are four grants $i_1 \succ i_2 \succ i_3 \succ i_4$, given by:

$$\begin{aligned} i_1 : \quad & b_{i_1} = [2, 2], \quad \rho(i_1) = \emptyset \\ i_2 : \quad & b_{i_2} = [1, 1], \quad \rho(i_2) = \{c_A\} \\ i_3 : \quad & b_{i_3} = [4, 4], \quad \rho(i_3) = \{c_A\} \\ i_4 : \quad & b_{i_4} = [2, 2], \quad \rho(i_4) = \{c_A\}, \end{aligned}$$

where $\mathcal{C} = \{c_A\}$, $\mathcal{B} = 4$, and $q_{c_A} = 3$. Consider the grant funding problem which omits the lowest merit grant i_4 . The unique allocation ξ satisfying non-wastefulness, respecting priorities, and satisfying strong feasible floors (Axioms A, B and C*) is $(2, 1, 0)$. It is worth seeing why $(2, 1, 0)$ satisfies both floor axioms before i_4 arrives. The category is short by $d_{c_A}(\xi) = 3 - 1 = 2$, and the only eligible grant that is not fully funded is i_3 , at a cost of 4. Since $4 > 2$, funding i_3 would overshoot the floor, so the category counts as satisfied under Axiom C and, i_3 being the only candidate, under Axiom C* as well. Adding i_4 changes nothing for Axiom C, whose threshold is still set by the highest-merit candidate i_3 ; but i_4 costs 2, so under Axiom C* the category is no longer satisfied, and closing the deficit forces i_1 out. In the problem which includes i_4 , the unique solution satisfying Axioms A, B, and C* is $(0, 1, 0, 2)$. Grant i_4 is non-marginal at $(2, 1, 0)$ —its minimum cost 2 exceeds the residual 1, and in its only category c_A the higher-merit grant i_3 is unfunded and so not fully funded—and it is non-contending there, since the funded grants i_1 and i_2 both outrank it. Therefore, we have exhibited a non-contending grant which affects the allocation. $\square$

Compare the allocation to the floors-first choice rule (which satisfies Axioms A, B and C). We have $C_{\Delta}^f(\mathcal{G}) = (2, 1, 0)$ when i_4 is omitted, and $C_{\Delta}^f(\mathcal{G}^{+i_4}) = (2, 1, 0, 0)$ when i_4 is included; the allocation is unaffected since C_{Δ}^f is independent of non-contending grants. The violation arises because strong feasible floors force the allocation to “squeeze in” additional funding for a category whenever doing so closes its floor deficit, even at the expense of higher-merit grants. The contrast between Axioms C* and C parallels that drawn by [Delacrétaz et al. \(2023\)](#) between envy-freeness, which rules out every priority violation, and interference-freeness, which rules out only those in which a higher-priority family could claim capacity used by the envied family (see Section 2). Like envy-freeness, Axiom C* tests the allocation against every eligible grant; like interference-freeness, Axiom C tests it only at the relevant margin.

Joint satisfiability

Whether Axioms A, B and C* can be met simultaneously in every floor problem is a question we leave open. For Axioms A–C, existence follows from the construction of C_{Δ}^f (Theorem 2), which advances an unsatisfied category by funding its highest-merit eligible grant that is not fully funded, a step that never creates a priority

violation. This argument does not extend to Axiom C*, under which a floor is met only once $d_c(\xi)$ falls below the minimum cost of every grant in $A_c(\xi)$; the threshold is thus set by the cheapest such grant, which may sit anywhere in the merit order. Meeting it may require funding a grant outranked under $\succ_\rho$ by an unfunded eligible grant, which Axiom B forbids once the funds committed reach that grant's minimum cost, or whenever an outranking grant is funded short of its ceiling. The proof of Proposition 10 illustrates both the tension and its limits. Axiom B blocks meeting the threshold through i_3 , since funding i_3 would leave i_2 unfunded with $\xi_{i_3} \geq \underline{b}_{i_2}$; the funds can instead be drawn from i_1 , which is incomparable under $\succ_\rho$ to every grant eligible for c_A because $\rho(i_1) = \emptyset$. A counterexample to joint satisfiability would have to close this route as well.

We have not been able to derive an impossibility from this tension. We instead report the results of an exhaustive computational search for counterexamples.⁵ Fixing a number of grants, a number of categories and a budget, the search enumerates every integer floor problem—cost bounds $1 \leq \underline{b}_i \leq \bar{b}_i \leq \mathcal{B}$, eligibility sets ranging over all subsets of $\mathcal{C}$, and floors $q_c \geq 1$ with $\sum_c q_c \leq \mathcal{B}$ —and for each enumerates every integer allocation, testing Axioms A, B and C* directly against their statements in Sections 4.3 and 4.4; for Axiom C* this means checking, at every category carrying a positive shortfall, that every grant in $A_c(\xi)$ is unfunded and costs more than $d_c(\xi)$. A problem is flagged if and only if no integer allocation satisfies all three axioms. Grants are indexed by merit position, $i_1 \succ \dots \succ i_n$, and cost bounds and eligibility are assigned to each position independently. Since the axioms refer to grants only through their data and the merit order, every problem of these dimensions is a relabeling of exactly one problem in this class, so fixing the order entails no loss of generality, and the counts below are of problems up to relabeling.

No problem was flagged. Across all fourteen configurations $(|I|, |\mathcal{C}|, \mathcal{B})$ — $(2, 1, 10)$, $(2, 2, 9)$, $(2, 3, 8)$, $(3, 1, 8)$, $(3, 2, 8)$, $(3, 3, 7)$, $(4, 1, 7)$, $(4, 2, 6)$, $(4, 3, 5)$, $(5, 1, 5)$, $(5, 2, 4)$, $(5, 3, 3)$, $(6, 1, 4)$ and $(6, 2, 3)$, approximately 5.2×10^9 problems up to relabeling of grants—every problem admits at least one allocation satisfying all three axioms, as does each of twelve million problems drawn at random with up to nine grants, four categories and budgets up to forty. In these draws the number of grants, the budget and the number of categories are uniform on $\{2, \dots, 9\}$, $\{2, \dots, 40\}$ and $\{1, \dots, \min(4, \mathcal{B})\}$; cost bounds and eligibility are drawn independently for each merit position; and the floors are uniform over $\{q : q_c \geq 1, \sum_c q_c \leq \mathcal{B}\}$. At these budgets, exhaustive enumeration of allocations is computationally prohibitive, so the allocations of each drawn problem are instead searched depth-first over the same space: grants are assigned integer funding levels one at a time in merit order, a branch is abandoned only

⁵The search code, together with the configurations reported below, is available at <https://github.com/joejourden/GE-B-R-R-R2/tree/main/code>.

once its partial allocation exceeds the budget, each completed allocation is tested against Axioms A, B and C* exactly as in the exhaustive search, and the search stops at the first allocation that passes. The only pruning discards vectors that are not allocations, so a drawn problem is flagged if and only if it has no integer allocation satisfying all three axioms. Each drawn problem is therefore resolved exactly, as in the exhaustive search; randomization enters only through the selection of problems, not through the search over allocations. We note two further properties of the search. First, the axioms are invariant to a common rescaling of the budget, the cost bounds and the floors, so the class of integer floor problems covers every problem of these dimensions whose data are commensurable and whose rescaled budget falls within the stated bound. Second, since an integer allocation satisfying the axioms establishes existence, restricting the search to integer allocations could only produce spurious flags, and none arose.

The configurations searched are small relative to the problems a funding body confronts, but they include the scale at which the other defects of strong feasible floors already appear: the failure of independence in Proposition 10 requires only four grants and a single category, whereas no instance of non-existence occurs anywhere in the range searched. Our case against Axiom C* therefore rests on its failure of independence rather than on any claim that the axioms are jointly unsatisfiable.

B NIH Review Process

As stated by the National Institutes of Health ([NIH, 2021](#)), after grant applications have been matched to appropriate reviewers, the funding decisions are made in two steps: (1) peer review, and (2) advisory council or board review that we summarize as follows:

(1) Peer review:

- (i) Initial peer review meetings are administered by either the Center for Scientific Review (CSR) or one of the NIH ICs (Institutes/Centers) with the funding authority as specified in the funding opportunity announcement (FOA).
- (ii) Reviewers provide an overall impact score to “reflect their assessment of the likelihood for the project to exert a sustained, powerful influence on the research field(s) involved, in consideration of the following review criteria, and additional review criteria (as applicable for the project proposed): significance, investigator(s), innovation, approach, environment” (as well

as additional unscored criteria). This score ranges from 1 (exceptional) to 9 (poor).

- (iii) The final overall impact score for each discussed application is determined by calculating the mean score from all the eligible members' final impact scores, and multiplying the average by 10.

(2) Advisory council or board

- (i) Members are chosen by the respective IC and are approved by the Department of Health and Human Services. For certain committees, members are appointed by the President of the United States.
- (ii) NIH program staff members examine applications and consider the overall impact scores given during the peer review process, percentile rankings (if applicable) and the summary statements in light of the Institute/Center's priorities.
- (iii) The Advisory Council/Board also considers the Institute/Center's goals and needs and advises the Institute/Center director concerning funding decisions.
- (iv) The Institute/Center director makes final funding decisions based on staff and Advisory Council/Board advice.

In short, in the first step, peer review assigns an overall impact score to each discussed application, and in the second step, the advisory council or board determines funding advice based on impact scores with consideration of institute objectives.

These scores are coarser than the strict merit order of our model. Because the overall impact score is a scaled mean of reviewers' integer scores, two discussed applications may receive the same score, and applications that are not discussed receive no overall impact score at all. The model's merit order $\succ$ is therefore not read off directly from this process; it is a strict ranking that extends the scores to all applications, discussed or not, and it is maintained as an assumption (see the discussion following Theorem 1).

Per the *NIH-Wide Strategic Plan for Fiscal Years 2016–2020*, for these years the NIH had the following objectives for HIV/AIDS research, which we quote directly (NIH, 2015):

HIV/AIDS Research Priorities

- Reduce incidence, including vaccines
- Safer, easier-to-use therapies
- Work towards a cure

- HIV-associated comorbidities, co-infections
- Cross-cutting areas: basic research, health disparities, training